

Multifractal Cryptocurrencies^{*}

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Abstract

We propose a multifractal model to capture the high frequency dynamics of cryptocurrency returns and show that our model accurately predicts cryptocurrency returns at five-minute intervals. We then introduce a novel trading strategy which provides trading signals based on risk-return trade-off. When combined with our model, the strategy generates substantial returns even under high trading fees, execution delays, and cryptocurrency crashes. We also document that the number of multifractal volatility states correlates with the cryptocurrency's age and wealth concentration. Finally, we present evidence of self-similarity in cryptocurrency volatility patterns across frequencies, reinforcing the multifractal dynamics on cryptocurrency returns.

JEL classification: C53, C58, E47, G11, G12, G17

Keywords: Cryptocurrency, Trading, Markov-switching multifractal, Approximate Maximum Likelihood

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1 Introduction

Following the 2024 U.S. elections, cryptocurrencies reached unprecedented highs and exhibited strong intraday swings, partly driven by policy-related commentary from the Trump administration. The high frequency swings were amplified by a predominantly young investor demographic connected via social platforms like Discord and Telegram, as well as the rapid adoption of crypto-sniping bots on decentralized exchanges (DEXs). As a result, price fluctuations displayed varying persistence and regime shifts at high frequency. The growing use of cryptocurrencies for everyday transactions, including remittances and payments across AI agents, further motivates an in-depth examination of their high frequency return behavior.

In this paper, we propose a model to capture five-minute cryptocurrency returns, allowing volatility to follow a variant of the Markov-switching multifractal (MSM) process introduced by [Calvet and Fisher \(2001\)](#) and proved to successfully capture multifrequency news in stock returns in [Calvet and Fisher \(2007\)](#). Our modeling choice is guided by two stylized facts of cryptocurrency markets. First, high frequency cryptocurrency prices exhibit strong volatility clustering, reflecting speculative trading flows, retail participants’ behavioral biases, and the layered activity of both short-term traders and long-term holders.¹ By incorporating multiple latent volatility components evolving at distinct timescales, the MSM structure naturally captures these overlapping effects. Second, the tail behavior of high frequency cryptocurrency returns often features extreme outliers.² By assigning distinct shocks to separate latent components, the MSM process naturally accommodates these spikes, offering enhanced flexibility compared to standard long-memory volatility models. We estimate the proposed MSM model via Approximate Maximum Likelihood (AML) introduced by [Czellar et al. \(2022\)](#).

¹Volatility clustering is often attributed to factors such as investor inertia and behavioral biases (see, e.g., [Teyssi re and Kirman 2006](#) and [Schmitt and Westerhoff 2017](#)).

²See, e.g., [Biais et al., 2023](#), and [Hackethal et al., 2022](#).

Our model contains dynamic mean and volatility processes, and captures market fluctuations and turbulence at a five-minute frequency. We compare our model’s performance with GARCH-type alternatives of our model.³ In sample, our model matches better empirical moments such as mean, daily momentum, standard deviations, kurtosis, and the decay of autocorrelation in volatility. Out of sample, our model generates more accurate forecasted returns than the benchmark models.

To evaluate the economic value of our model’s forecasting performance, we introduce a novel trading strategy based on Dynamic Sharpe Ratio (DSR) signals, which are based on a risk-return trade-off. The idea is to compare the ratio of the forecasted mean to the forecasted volatility at each five-minute interval, going long or short if that ratio surpasses a threshold. When implementing the strategy to align with cryptocurrency trading platforms like Binance and Kraken, we factor in both the position opening fees (which include the bid-ask spread) and an hourly rollover fee for maintaining long or short positions. When the DSR strategy is applied to a portfolio of the top five cryptocurrencies with signals from the MSM model, the trading strategy generates an 805% return with annualized Sharpe ratios of 3.05 over a two-year out-of-sample period. Over the same interval, a buy-and-hold trading strategy yields -61%. To ensure robustness, we also analyze various opening fees and execution delays. Even with position opening fees as high as 0.2% and trade execution delays of 150 seconds per position switch, the MSM-based DSR strategy continues to deliver positive returns.

To isolate sources of profitability, we decompose returns into contributions from long and short positions. Notably, the MSM-based DSR signals yield fairly balanced gains in both directions, reflecting the model’s ability to respond to upward and downward price swings. We also compare strategy performance on *crash days*, e.g. when the cryptocurrency prices drop more than 10% in a day, versus stable days. The MSM-based DSR signals consistently

³Models that focus solely on volatility, such as the HAR model by [Corsi \(2009\)](#), are not applicable here because they do not account for the mean process of cryptocurrency returns. Moreover, since we model and forecast returns at the highest available frequency of five minutes, realized volatility is not available.

generate positive returns under both downward spikes and more placid market conditions.

We further investigate which cryptocurrency characteristics potentially drive multifractal parameter estimates. In particular, we find that the number of latent volatility states correlates positively with coin age and the fraction of wealthy coin holders. This suggests that greater volatility persistence may coincide with a more mature user base and a concentrated wealth distribution. Finally, we assess self-similarity in high frequency cryptocurrency data by comparing parameter estimates of returns captured at five-minute, ten-minute, and fifteen-minute intervals. Our results show that the number of latent states remains invariant across these frequencies, supporting fractal nature of high frequency cryptocurrency price fluctuations.

We contribute to the growing literature on cryptocurrencies along two dimensions. First, the literature on the behavior of cryptocurrency holders should be helpful in understanding the return and volatility dynamics of the market. The previous studies identify various drivers that contribute to the complex return and volatility patterns in the cryptocurrency market: manipulation (see e.g., [Griffin and Shams 2020](#): stablecoin manipulation, [Cong et al. 2023](#) and [Aloosh and Li 2024](#) : wash trading), the dominance of retail investors (see e.g., [Pursiainen and Toczynski 2022](#): cryptocurrency retail demographics, [Divakaruni and Zimmerman 2024](#): stimulus checks and Bitcoin price), the behavioral patterns of crypto investors (see e.g., [Hackethal et al. 2022](#): media sentiment over investors, [Chava et al. 2025](#): gambling behavior patterns), changes in market sentiment (see e.g., [Biais et al. 2023](#): beliefs on transactional benefit), introduction of new paradigms (see e.g., [Nimalendran et al. 2025](#): market efficiency over IDOs), and interconnected communities (see e.g., [Li et al. 2025](#): pump-and-dump schemes via Telegram channels). Fewer papers investigate extreme-volatility aspects of cryptocurrency through a multifractal lens. Our findings show that the layered volatility states in MSM reflect precisely these behavioral and structural influences.

Second, we also provide new insights to the literature that applies established finance tools to cryptocurrency markets. In [Fieberg et al. \(2025\)](#), [Liu et al. \(2022\)](#), and [Liu and](#)

Tsyvinski (2021), factor models are used to capture daily cryptocurrency returns, paralleling conventional asset pricing approaches. Lee and Wang (2025) use realized variance to predict weekly returns of cryptocurrencies. Makarov and Schoar (2020) analyze the signed volume of trades to explain Bitcoin return variations across five-minute to daily intervals. Kogan et al. (2024) show that retail traders use adoption-based momentum-like strategies in trading Bitcoin, Ethereum, and Ripple. We complement these studies by offering a tractable model that encapsulates cryptocurrency return dynamics and proposing a practical trading strategy that links the volatility theory with real-time cryptocurrency trading.

The remainder of this paper is organized as follows. Section 2 describes the data and the summary statistics. Section 3 presents the MSM model for cryptocurrency returns and the AML method to estimate model parameters, and discusses the implications of cryptocurrency network characteristics for volatility persistence, as well as the self-similarity in high frequency cryptocurrency returns. Section 4 compares the out-of-sample performance of the MSM model with benchmark models. Section 5 introduces the trading with DSR signals and gauges its performance under varying trading fee and execution delays. Section 6 concludes, and further details are provided in the appendix.

2 The Data

We collect price data (in US dollars) of major cryptocurrencies at the five-minute frequency via CoinMarketCap, a widely recognized source for cryptocurrency price and volume data, aggregating information from over 250 major exchanges. CoinMarketCap offers comprehensive five-minute interval data on opening and closing prices.⁴ The price is calculated as the volume-weighted average of market pair prices in all cryptocurrency exchanges.⁵ The sample

⁴CoinMarketCap data is widely used in the literature (see Liu et al., 2022). See also the API documentation: <https://coinmarketcap.com/api/documentation/v1>.

⁵Aggregating data across multiple exchanges enhances the robustness of our price measure by mitigating issues such as non-free market pricing due to disabled withdrawals or deposits on any single platform. This approach also minimizes the impact of isolated outlier prices caused by technical glitches, as CoinMarketCap’s algorithm systematically excludes these anomalies. For a detailed explanation of the price calculation, see <https://support.coinmarketcap.com/hc/en-us/articles/>

spans from September 1, 2016, 00:04 GMT until August 31, 2023, 23:59 GMT. We focus on the five largest cryptocurrencies, based on their market capitalization at the start of our sample - Bitcoin, Dash, Ethereum, Litecoin, and Ripple - whose combined market capitalization represented approximately 92% of the total cryptocurrency market capitalization on September 1, 2016.

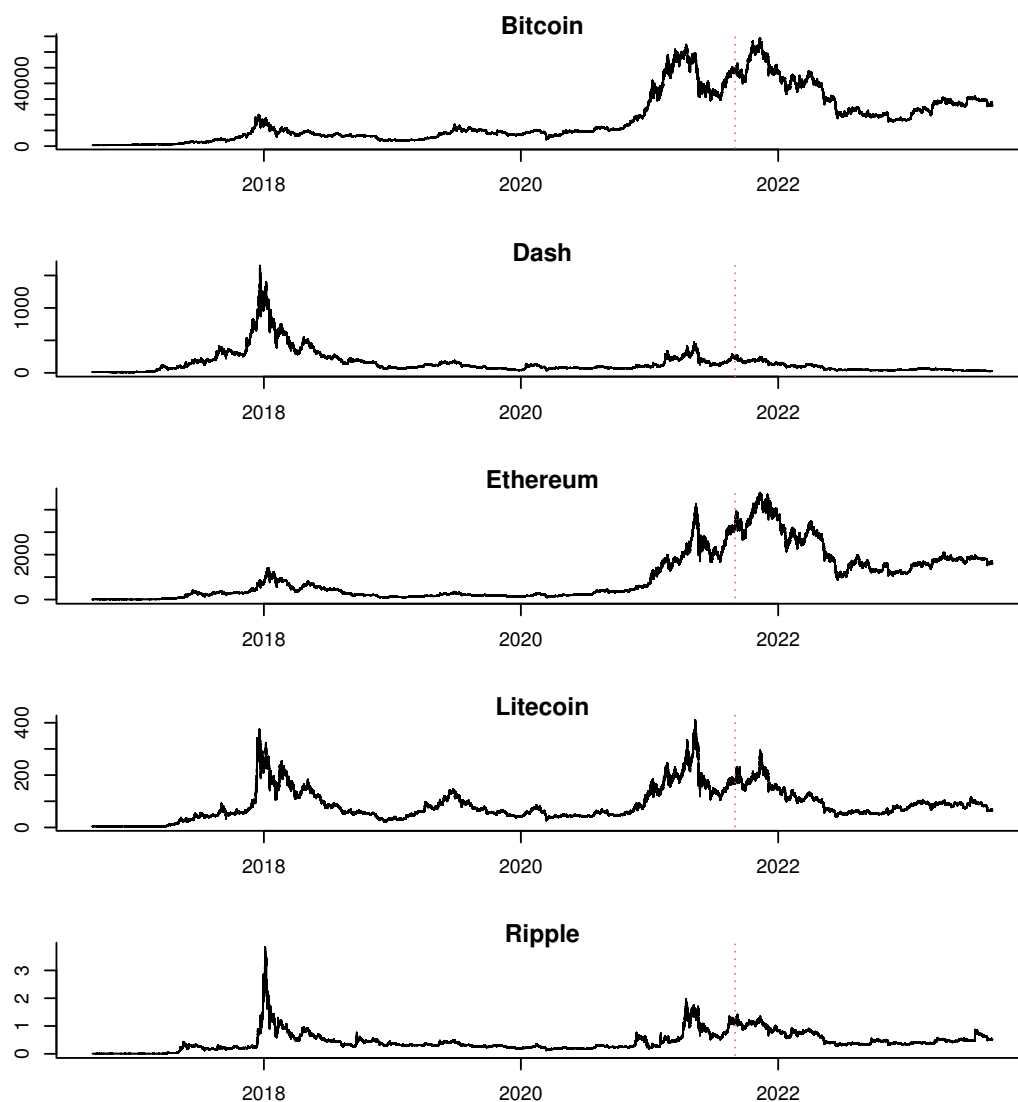


Fig. 1. *Price Series of Cryptocurrencies.* Notes: This figure illustrates the cryptocurrency prices at the five-minute frequency from September 1, 2016, 00:04 GMT to August 31, 2023, 23:59 GMT. The price is defined in USD per coin unit. The red dotted line marks the split between the in-sample and out-of-sample periods.

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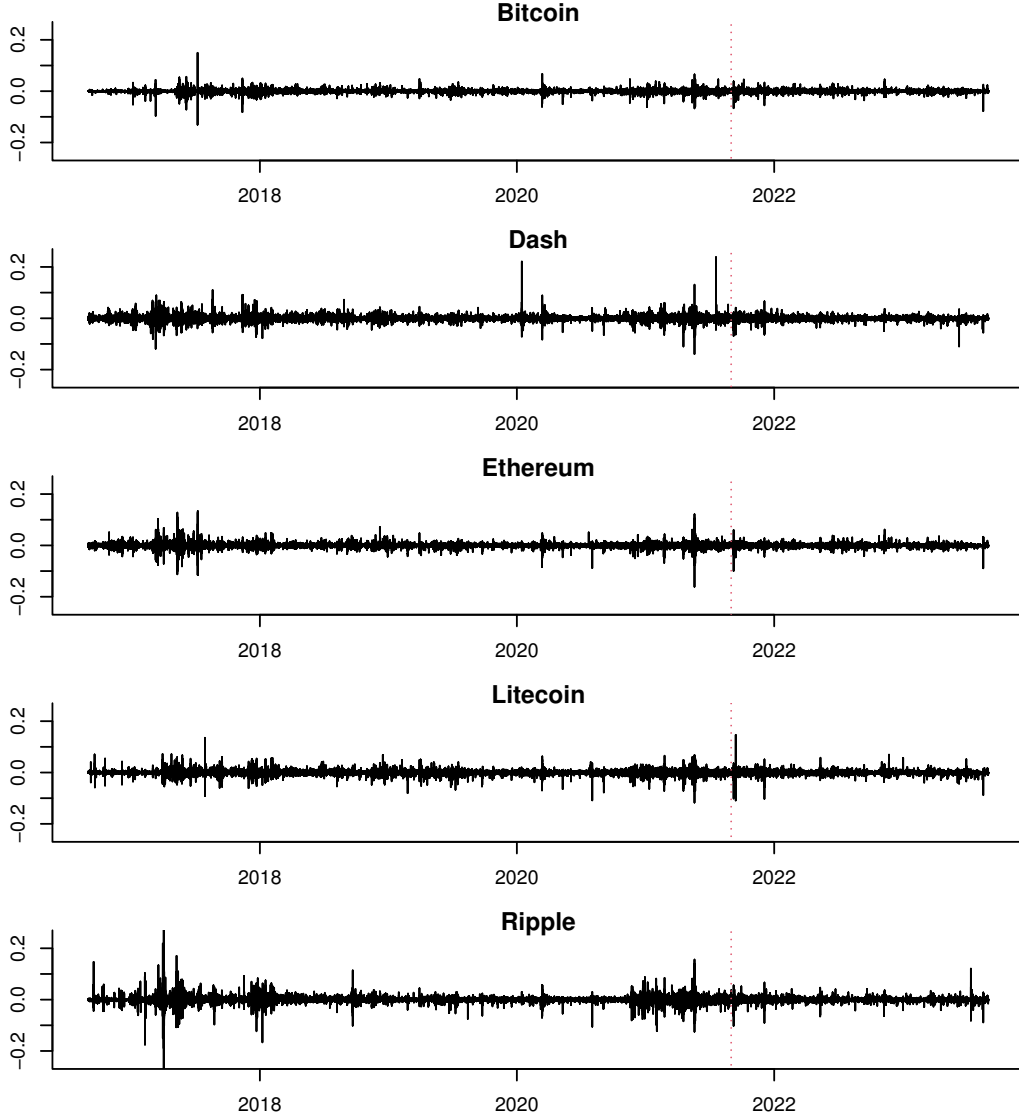


Fig. 2. *Return Series of Cryptocurrencies.* Notes: This figure illustrates five-minute cryptocurrency returns from September 1, 2016, 00:04 GMT to August 31, 2023, 23:59 GMT. The red dotted line marks the split between the in-sample and out-of-sample period.

Figure 1 depicts the price series of the cryptocurrencies. We use the five-minute prices to construct simple cryptocurrency returns. Figure 2 depicts the return series of cryptocurrencies. We divide the sample into an in-sample period from September 1, 2016, 00:04 GMT to August 31, 2021, 23:59 GMT and an out-of-sample period from September 1, 2021, 00:04 GMT to August 31, 2023, 23:59 GMT. We denote $T = 525,888$ the last in-sample date and $T' = 736,128$ the last out-of-sample date. The end of the in-sample period T is marked with vertical dotted lines in Figures 1 and 2.

Table 1 reports the summary statistics of the cryptocurrency returns $r_{1:T'}$. Given that cryptocurrency returns often exhibit large outliers and that higher order moments can be unstable under such conditions, we employ robust measures of skewness and kurtosis.⁶ The sample means are all positive, indicating a long-term upward trend. The sample standard deviations are very close to one another. Skewness is not typically negative as we use simple returns⁷. Excess kurtosis is positive as expected.

Since we are interested in dynamically capturing the behaviour of cryptocurrency returns, in order to understand how past returns influence future returns, we report autocorrelations of returns and absolute returns in Table 2. We refer to the first-order autocorrelation in returns as Momentum (denoted Mom in Table 2). Momentum is persistently positive for all five return series indicating the existence of a short-term trend in major cryptocurrencies. Next, we investigate whether there exists a longer term trend. Figure 3 plots the correlation of returns r_t with mean lagged returns $\overline{r_{(t-n):(t-1)}}$ for $n \geq 1$. In all five series, we first see a sharp decay in the autocorrelations, then a sudden rise that reaches a maximum around $n = 288$, then a decay again. Since one day contains 288 five-minute periods, we conclude that five-minute cryptocurrency returns r_t are influenced by daily mean lagged returns $\overline{r_{(t-288):(t-1)}}$. We therefore report in Table 2 the autocorrelations of returns r_t with daily mean lagged returns $\overline{r_{(t-288):(t-1)}}$, which we label Daily Momentum (denoted Mom_D in Table 2).

We then investigate the dynamic behavior of volatility. The last three columns of Table 2 present the autocorrelations in absolute returns at lags one, two, and three, denoted AC_1 , AC_2 and AC_3 , respectively⁸. Autocorrelations in absolute returns are very large and

⁶Concretely, we use the robust skewness measure of Kim and White (2004):

$$\text{Skewness} = \frac{\text{Mean}\{r_{1:T'}\} - \text{Median}\{r_{1:T'}\}}{\text{St. dev.}\{r_{1:T'}\}},$$

and the robust excess kurtosis of Hogg (1972):

$$\text{Excess kurtosis} = \frac{U_{0.05} - L_{0.05}}{U_{0.5} - L_{0.5}} - 2.585,$$

where $U_\alpha = \text{Mean}\{r_t \in r_{1:T'} \mid r_t > q_{1-\alpha}\}$, $L_\alpha = \text{Mean}\{r_t \in r_{1:T'} \mid r_t < q_\alpha\}$ and q_β is the β^{th} quantile of $r_{1:T'}$.

⁷Log returns would add negative skewness to the data.

⁸To measure persistence in volatility, we use autocorrelations in absolute returns which are more stable

of similar size for the five return series and exhibit a slow decay. The rate of decay in autocorrelations varies across cryptocurrencies. For example, Ripple shows a substantially slower decay than other cryptocurrencies.

Table 1
Basic Statistics of Cryptocurrency Returns

	Ticker	Mean $\times 10^4$	S.D. $\times 10^4$	Skew.	Exc. Kurt.
Bitcoin	BTC	0.071	19.625	-0.000	1.697
Dash	DASH	0.059	31.158	0.001	1.570
Ethereum	ETH	0.100	25.491	0.003	1.600
Litecoin	LTC	0.079	28.402	0.003	1.559
Ripple	XRP	0.128	36.748	0.003	2.052

This table presents the basic statistics (mean, standard deviation, robust skewness, and robust excess kurtosis) for the five-minute cryptocurrency returns from September 1, 2016, 00:04 GMT to August 31, 2023, 23:59 GMT.

Table 2
Momentum, Daily Momentum, and Autocorrelations in Absolute Returns

	Ticker	Mom	Mom _D	AC_1	AC_2	AC_3
Bitcoin	BTC	0.102	0.013	0.447	0.357	0.339
Dash	DASH	0.085	0.008	0.413	0.350	0.328
Ethereum	ETH	0.133	0.015	0.437	0.355	0.333
Litecoin	LTC	0.129	0.013	0.412	0.359	0.334
Ripple	XRP	0.062	0.012	0.463	0.426	0.412

This table presents momentum (Mom), daily momentum (Mom_D), and autocorrelations in absolute returns (AC_1 , AC_2 , AC_3) for the five-minute cryptocurrency returns from September 1, 2016, 00:04 GMT to August 31, 2023, 23:59 GMT.

in the presence of outliers than autocorrelations in squared returns.

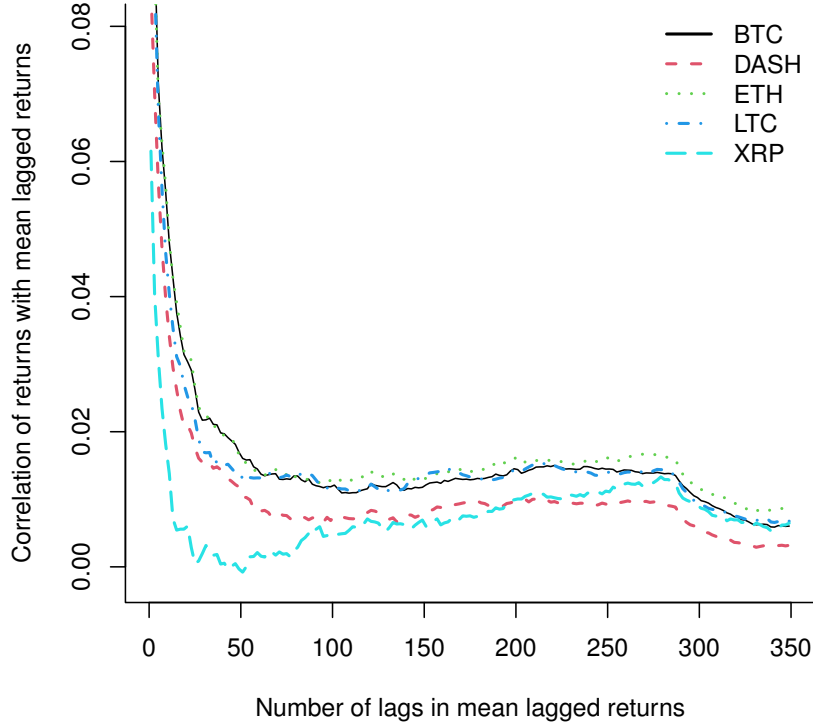


Fig. 3. *Trends in Return Series of Cryptocurrencies.* Notes: This figure illustrates correlations of five-minute returns r_t , with mean lagged returns $\bar{r}_{(t-n):(t-1)}$ for $n \geq 1$. The x-axis represents the number of lags (n -period). Correlation is calculated for the period from September 1, 2016, 00:04 GMT to August 31, 2023, 23:59 GMT.

To summarize, mean cryptocurrency returns exhibit long-term, daily and five-minute trends. Volatility is highly persistent, although persistence levels vary across cryptocurrencies. Finally, very large outliers can occasionally occur. The following section introduces a multifractal model capable of generating returns with long-term, daily and five-minute trends, persistent volatilities with adjustable levels, and large kurtosis.

3 The Model

In this section, we first describe a multifractal model that captures both mean and volatility characteristics of high frequency cryptocurrency returns. Second, we estimate the model via AML. Finally, we discuss the multifractal characteristics of the cryptocurrency markets.

3.1 A Markov-Switching Multifractal Process

We propose a model that captures the long-term, daily, and five-minute trends, as well as kurtosis and volatility persistence levels of cryptocurrency returns. We assume that cryptocurrency returns evolve according to

$$r_t = \mu_t + \sigma_t u_t \quad (3.1)$$

with

$$\mu_t = \mu + \nu \overline{r_{(t-D):(t-1)}} + \rho r_{t-1}, \quad (3.2)$$

$$\sigma_t^2 = \sigma^2 \prod_{k=1}^{\bar{k}} M_{k,t}, \quad (3.3)$$

for $t > D$ with $D = 288$ and $\mu_t = \mu/(1 - \nu - \rho)$ for $t \leq D$. The error terms $\{u_t\}_{t=1}^T$ are independent and identically distributed standard normal variables. The volatility σ_t follows a Binomial Markov-Switching Multifractal process (Calvet and Fisher 2001, 2007, 2008) defined in the following way. Each latent $M_{k,t}$ component can take two possible values, m_0 and $2 - m_0$ with $1 < m_0 < 2$. Consequently, the latent state vector $M_t = (M_{1,t}, \dots, M_{\bar{k},t})'$ can take $d = 2^{\bar{k}}$ possible values m^j , $j = 1, \dots, d$, and evolves according to the following rule. Given a value $M_{k,t}$ for the k^{th} component at time $t \geq 1$, the next-period multiplier is drawn from

$$M_{k,t+1} = \begin{cases} m_0^{B_{k,t}} (2 - m_0)^{1-B_{k,t}} & \text{with probability } \gamma_k \\ M_{k,t} & \text{with probability } (1 - \gamma_k) \end{cases}, \quad (3.4)$$

where $\{B_{k,t}\}$ are independent and identically distributed Bernoulli(0.5) variables. The initial state M_0 is generated from the stationary distribution λ_0 defined as

$$\lambda_0^j = \mathbb{P}[M_0 = m^j] = 1/d, \quad \forall j = 1, \dots, d. \quad (3.5)$$

Furthermore, the probabilities $\gamma_k, k = 1, \dots, \bar{k}$, are parameterized such that the first component of $M_{1,t}$ is the most persistent one, while the last one $M_{\bar{k},t}$ is the least persistent one. More specifically:

$$\gamma_k = \bar{\gamma} b^{k-\bar{k}}, \quad \bar{\gamma} \in (0, 1], \quad b > 1, \quad k = 1, \dots, \bar{k}. \quad (3.6)$$

Hereafter, we refer to the model defined by equations (3.1)-(3.6) as the MSM model. The MSM model contains eight parameters, each of them capturing a specific feature of the cryptocurrency return dynamics. The parameter μ captures the long-term trend in the mean cryptocurrency return, ν the daily trend, and ρ the five-minute trend. The parameter m_0 can generate an appropriate kurtosis, $\bar{\gamma}$ the first order autocorrelation in absolute returns (AC_1), b the speed of decay in the autocorrelations ($AC_1 - AC_2$), σ the long term volatility, and $\bar{k}$ the persistence level in the volatility. In Figure 4, we illustrate the role of each parameter. More precisely, we simulate return series of size $T = 525,888$ from the model defined in (3.1)-(3.3) with parameters $\mu = 0.00001$, $\nu = 0.05$, $\rho = 0.1$, $m_0 = 1.4$, $\bar{\gamma} = 0.9$, $b = 1.7$, $\sigma = 0.002$, and $\bar{k} = 10$, and report eight summary statistics captured by the eight parameters. In each panel of Figure 4, one parameter varies while the others remain fixed.

The top left panel shows that the sample mean of the simulated series increases when the value of μ increases. The top right panel indicates that the daily momentum (Mom_D) increases when ν increases. The second left panel shows that momentum (Mom) is an increasing function of ρ . The second right panel illustrates that excess kurtosis increases when m_0 increases. The third left panel shows that the first autocorrelation of absolute returns (AC_1) decreases when $\bar{\gamma}$ increases. The third right panel shows that the speed of decay in the autocorrelations of absolute returns ($AC_1 - AC_2$) decreases when b increases. The bottom left panel shows that the standard deviation of returns increases with σ , and the bottom right panel shows that the half-life of autocorrelations in absolute returns⁹ (Half-life AC) increases with $\bar{k}$.

⁹Half-life of autocorrelations in absolute returns is defined as the smallest integer h such that $AC_h \leq AC_1/2$.

Denote the structural parameter vector

$$\theta = (\mu, \nu, \rho, m_0, \bar{\gamma}, b, \sigma, \bar{k})'.$$

The log-likelihood associated with the observed returns $r_{1:T}$ is given by:

$$L(\theta|r_{1:T}) = \frac{1}{T} \sum_{t=1}^T \log \left[\sum_{j=1}^d f(r_t|M_t = m^j)(\lambda_{t-1}A)_j \right], \quad (3.7)$$

where $f(r_t|M_t = m^j)$ is the probability density function evaluated at r_t for a normal distribution $\mathcal{N}[\mu_t; \sigma^2 \Pi_{k=1}^{\bar{k}} m_k^j]$. The conditional probabilities $\lambda_t^j = \mathbb{P}[M_t = m^j|r_{1:t}]$, $j = 1, \dots, d$, are computed recursively:

$$\lambda_t = \frac{f(r_t) \odot (\lambda_{t-1}A)}{f(r_t)(\lambda_{t-1}A)^\top}, \quad t \geq 1, \quad (3.8)$$

where λ_0 is given in (3.5), $f(r_t)$ is the vector of conditional densities $(f(r_t|M_t = m^j))_{1 \leq j \leq d}$, and $\odot$ denotes the Hadamard product $x \odot y = (x_1 y_1, \dots, x_d y_d)$ for all $x, y \in \mathbb{R}^d$. The transition matrix $A = (a_{i,j})_{1 \leq i,j \leq d}$ satisfies

$$a_{i,j} = \mathbb{P}[M_t = j | M_{t-1} = i] = \prod_{k=1}^{\bar{k}} \left[(1 - \gamma_k) 1_{\{m_k^i = m_k^j\}} + \frac{\gamma_k}{2} \right],$$

where $1_{\mathcal{A}}$ is equal to one if $\mathcal{A}$ is true and is zero otherwise.

The log-likelihood is available in closed form. However, estimating θ by maximum likelihood is challenging as a single evaluation requires $O(2^{2\bar{k}}T)$ computations, where $O(\cdot)$ stands for the order of the evaluation. If $\bar{k}$ is large, estimation via maximum likelihood becomes prohibitively expensive, especially in large sample sizes such as high frequency data. As an example, when $\bar{k} = 30$ and $T = 525,888$, it would take 131,654.7 years to evaluate the log-likelihood function and approximately 20,000 times more time to maximize it. The following section explains estimation via AML, which significantly reduces computation time.¹⁰

¹⁰A GMM approach to estimate σ and m_0 in an MSM model has also been introduced by Lux (2008). However, to our knowledge, AML is the only computationally feasible technique that estimates all five volatility parameters $(m_0, \bar{\gamma}, b, \sigma, \bar{k})$ in an MSM.

In particular, when $\bar{k} = 30$, evaluating the AML objective function only takes 23.5 seconds to compute.¹¹

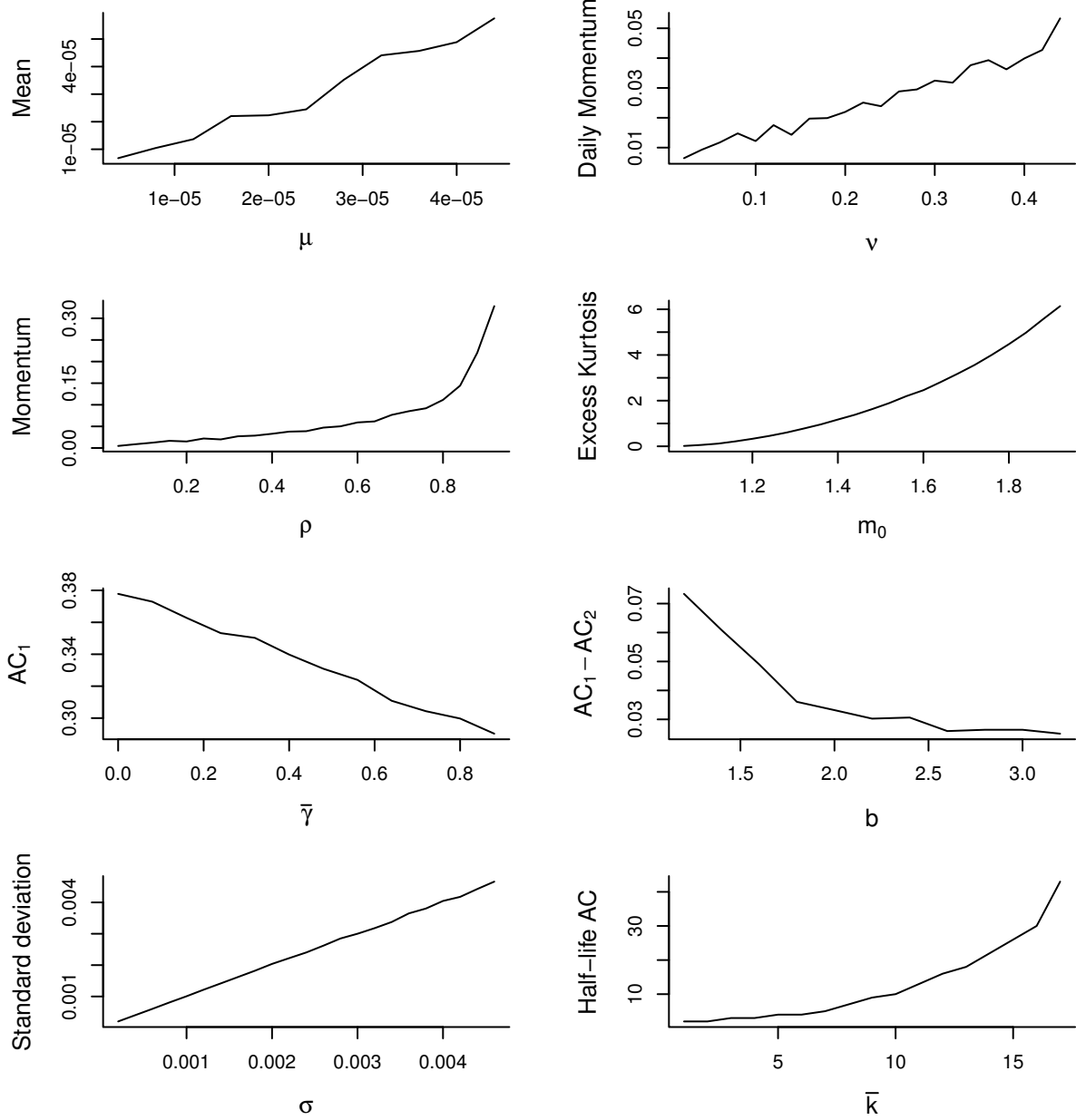


Fig. 4. *Simulated Moments.* Notes: This figure illustrates the role of each parameter in the MSM model. In each panel, the x-axis represents the varying parameter, while all other parameters are kept fixed. The y-axis represents the median of 100 summary statistics obtained with simulated MSM series. The simulation is based on return series of size $T = 525,888$ from the MSM model with parameters $\mu = 0.00001$, $\nu = 0.05$, $\rho = 0.1$, $m_0 = 1.4$, $\bar{\gamma} = 0.9$, $b = 1.7$, $\sigma = 0.002$, and $\bar{k} = 10$.

¹¹Programs were implemented in C (using the Intel compiler) and computation was performed on an Intel(R) Xeon(R) Gold 5218 CPU @ 2.30GHz.

3.2 Estimation via Approximate Maximum Likelihood

We partition the structural parameter vector as follows:

$$\theta = (\zeta', \bar{k})', \zeta = (\mu, \nu, \rho, m_0, \bar{\gamma}, b, \sigma)'.$$

At the constraint $\bar{k} = 2$, the log-likelihood $L(\zeta, 2|r_{1:T})$ has a computational cost that is only $O(2^4 T)$ which for $T = 525,888$ takes 0.05 seconds to run on a powerful server. Denote the constrained maximum likelihood estimator at $\bar{k} = 2$ by:

$$\hat{\zeta} = \arg \max_{\zeta} L(\zeta, 2|r_{1:T}). \quad (3.9)$$

Define a pseudo-score vector by

$$G(\zeta, 2|r_{1:T}) = \left[\frac{\partial L(\zeta, 2|r_{1:T})}{\partial \zeta'}, L([\zeta, 3|r_{1:T}) - L(\zeta, 2|r_{1:T})] \right]'.$$

Consider a pseudo-data series $\tilde{r}_{1:T}^{(h)}(\theta)$ (for $h = 1, \dots, H$) simulated from the structural model with parameter values $\theta \in \Theta_0$. The simulated pseudo-score vector calculated using the pseudo-data is:

$$G[\zeta, 2|\tilde{r}_{1:T}^{(h)}(\theta)] = \left[\frac{\partial L[\zeta, 2|\tilde{r}_{1:T}^{(h)}(\theta)]}{\partial \zeta'}, L[\zeta, 3|\tilde{r}_{1:T}^{(h)}(\theta)] - L[\zeta, 2|\tilde{r}_{1:T}^{(h)}(\theta)] \right]'.$$

The Approximate Maximum Likelihood (AML; [Czellar et al., 2022](#)) estimator $\hat{\theta}$ of θ is the solution to the optimization problem:

$$\hat{\theta} = \arg \min_{\theta} \left\| G[\hat{\zeta}, 2|r_{1:T}] - \frac{1}{H} \sum_{h=1}^H G[\hat{\zeta}, 2|\tilde{r}_{1:T}^{(h)}(\hat{\theta})] \right\|^2, \quad (3.10)$$

where $\|x\|$ denotes the Euclidean norm of x . [Czellar et al. \(2022\)](#) show that under regularity conditions, the AML estimator is consistent of the true data generating parameter θ^0 and is

asymptotically normally distributed:

$$\sqrt{T}(\hat{\theta} - \theta^0) \rightarrow_d \mathcal{N}(0, \Omega),$$

with $\Omega = (1 + H^{-1})J^{-1}I(J')^{-1}$ and

$$I = \lim_{T \rightarrow \infty} \text{Var} \left\{ \sqrt{T}G \left[\zeta^0, 2|\tilde{r}_{1:T}^{(1)}(\theta^0) \right] \right\}, \quad (3.11)$$

$$J = -\text{plim}_{T \rightarrow \infty} \frac{\partial G \left[\zeta^0, 2|\tilde{r}_{1:T}^{(1)}(\theta) \right]}{\partial \theta'} \bigg|_{\theta=\theta^0}, \quad (3.12)$$

where ζ^0 is the probability limit of $\hat{\zeta}^{12}$. In the empirical estimation, we use $H = 20$ number of pseudo-data series.¹³

3.3 Parameter Estimates

We estimate the MSM parameters using the in-sample returns. Table 3 presents the AML parameter estimates of the MSM model. All parameters are statistically significant at the 5% level, except for the long-term mean parameter μ for Dash, Litecoin, and Ripple, and daily momentum parameter ν for Dash and Ripple. Parameters μ and ν are the most significant in the case of Bitcoin which suggests that Bitcoin exhibits a strong inherent tendency towards

¹²In numerical computations we use

$$\hat{I} = \frac{1}{S-1} (G_d - \overline{G_d})(G_d - \overline{G_d})' \text{ with } G_d = \left(G \left[\hat{\zeta}, 2|\tilde{r}_{1:T}^{(1)}(\hat{\theta}) \right], \dots, G \left[\hat{\zeta}, 2|\tilde{r}_{1:T}^{(S)}(\hat{\theta}) \right] \right) \quad (3.13)$$

with $\overline{G_d} = \frac{1}{S} \sum_{s=1}^S G \left[\hat{\zeta}, 2|\tilde{r}_{1:T}^{(s)}(\hat{\theta}) \right]$ and

$$\hat{J} = \frac{1}{S} \sum_{s=1}^S \frac{\partial G \left[\hat{\zeta}, 2|\tilde{r}_{1:T}^{(s)}(\theta) \right]}{\partial \theta'} \bigg|_{\theta=\theta^0}, \quad (3.14)$$

where $\partial G[\hat{\zeta}, 2|\tilde{r}_{1:T}^{(s)}(\theta_{1:7}, \bar{k})]/\partial \bar{k}$ is approximated by $G[\hat{\zeta}, 2|\tilde{r}_{1:T}^{(s)}(\theta_{1:7}, \bar{k} + 1)] - G[\hat{\zeta}, 2|\tilde{r}_{1:T}^{(s)}(\theta_{1:7}, \bar{k})]$, as for the other parameters $\partial G[\hat{\zeta}, 2|\tilde{r}_{1:T}^{(s)}(\theta)]/\partial \theta_j$ is approximated by numerical differentiation. We use $S = 1,000$ in the standard error calculations of our empirical results.

¹³Czellar et al. (2022, Proposition 3) show that under regularity conditions, an AML estimator achieves the Cramer-Rao efficiency bound but the asymptotic variance is multiplied by $(1+1/H)$. Hence, a choice of $H = 20$ corresponds to an efficiency loss of 5% (as $1 + (1/20) = 1.05$).

positive long-term returns and a persistent relationship between its daily momentum and subsequent returns. The five-minute momentum parameter ρ remains significant and positive for all cryptocurrencies. All multifractal parameters are highly significant.

Table 3
Parameter Estimates of the MSM Model via AML

	$\mu \times 10^4$	$\nu \times 10$	$\rho \times 10$	m_0	$\bar{\gamma}$	b	$\sigma \times 10^2$	$\bar{k}$
BTC	0.113 (0.010)	0.668 (0.047)	1.182 (0.018)	1.360 (0.022)	0.901 (0.112)	1.751 (0.234)	0.201 (0.052)	23 (3.258)
DASH	0.048 (0.064)	0.106 (0.586)	0.270 (0.017)	1.582 (0.010)	0.921 (0.166)	5.745 (0.533)	0.396 (0.015)	6 (0.331)
ETH	0.076 (0.025)	0.462 (0.226)	1.176 (0.017)	1.501 (0.028)	0.400 (0.055)	3.501 (0.856)	0.339 (0.022)	8 (1.208)
LTC	-0.033 (0.057)	-0.899 (0.195)	0.418 (0.016)	1.323 (0.013)	0.954 (0.140)	1.721 (0.161)	0.354 (0.042)	21 (1.991)
XRP	-0.043 (0.157)	-0.500 (1.699)	0.220 (0.029)	1.390 (0.033)	0.477 (0.062)	2.052 (0.414)	0.457 (0.134)	18 (3.326)

This table presents parameter estimates of the MSM model via AML using five-minute cryptocurrency returns from September 1, 2016, 00:04 GMT to August 31, 2021, 23:59 GMT. Asymptotic standard errors are reported in parentheses.

3.4 Age and Holder Composition in Cryptocurrencies

A noteworthy parameter in MSM models is $\bar{k}$ which governs the depth of volatility persistence. Given the distinctive market structure of cryptocurrencies, we investigate whether $\bar{k}$ correlates with certain network-level characteristics, particularly those pertaining to the wealth distribution of coin holders and the cryptocurrency’s age.¹⁴ The rationale is twofold. First, large holders, not necessarily whales in the network, often play a disproportionately significant role in price dynamics by triggering or amplifying volatility through concentrated trades. Second, cryptocurrencies with well-established user bases and longevity may exhibit more persistent volatility patterns, as the trading behavior of experienced or well-capitalized

¹⁴We define the cryptocurrency’s age starting from day zero, marked by the first recorded blockchain event (e.g., genesis block or the first block after a fork).

participants tends to be more systematic and of longer-horizon in nature.

To test these ideas, we collect daily network data from coinmetrics.io, a prominent repository of on-chain metrics used in the recent literature (see e.g. [Ranaldo et al. 2024](#)). The dataset spans various indicators, including the number of active addresses, transaction fees, and mining statistics. We focus on the *proportion of wealthy coin holders*. Specifically, we define this term as the share of addresses holding more than 100 USD worth of a native cryptocurrency relative to the total number of addresses with a positive balance.¹⁵ Figure 5 illustrates the relationship between $\bar{k}$ and two attributes: the age of each cryptocurrency as of the start of our in-sample period (left panel), and the daily average proportion of wealthy coin holders over the in-sample period (right panel).

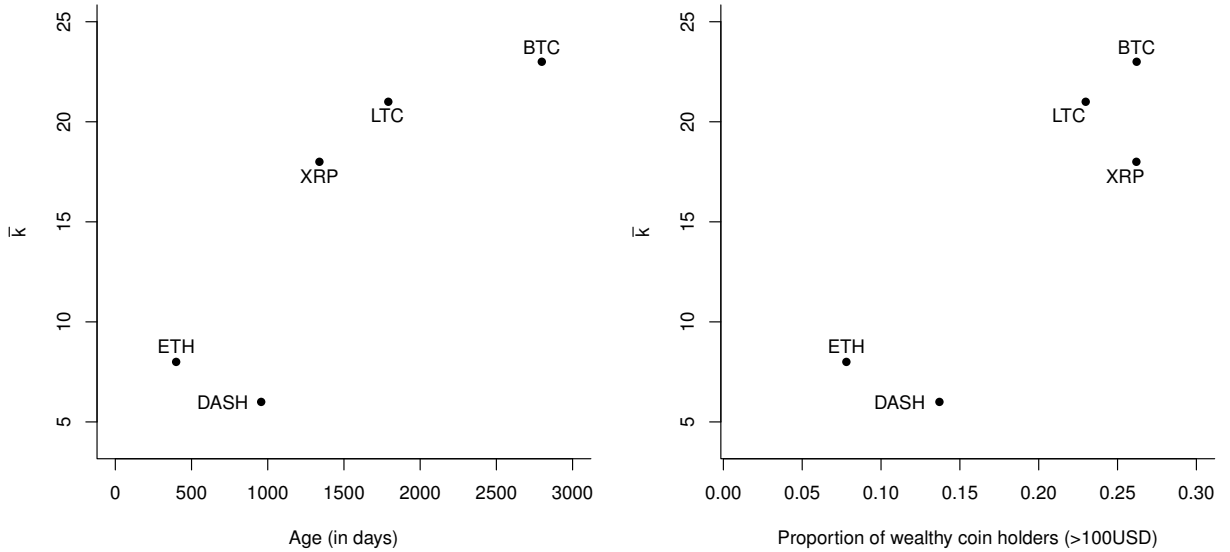


Fig. 5. *Cryptocurrency Metrics and the Number of Volatility States $\bar{k}$* . Notes: The age of a cryptocurrency (left panel) and the proportion of wealthy coin holders (with more than 100 USD native token balance) (right panel) are plotted with respect to the number of volatility states $\bar{k}$. The age of a cryptocurrency is calculated as of September 1, 2016, the first day of in-sample period. The daily average of proportion of wealthy coin holders is calculated for the period September 1, 2016 - August 31, 2021.

Across the remaining currencies, a consistent positive relationship emerges. Those with

¹⁵Native cryptocurrency (coin) is defined as the inherent coin used in a network to pay for transaction fees, issue mining rewards, and make transfers from one account to another. For example, Ether (ETH) is the native coin of the Ethereum network, which concurrently supports alternative token standards like ERC-20, ERC-721, and ERC-1155, each tailored for specific functionalities within the ecosystem.

longer market histories and higher shares of wealth-concentrated addresses exhibit higher $\bar{k}$ estimates, implying a large volatility persistence. Intuitively, longer-established markets tend to develop a user base that includes significant stakeholders whose strategic trading can induce persistence in volatility. Similarly, a higher fraction of wealthy coin holders may magnify and propagate shocks over multiple time scales, further boosting the number of volatility states $\bar{k}$. Overall, these results support that $\bar{k}$ encapsulates deeper structural features of a cryptocurrency market, such as investor composition and market maturity.

3.5 Self-similarity in Cryptocurrencies

To test the notion of self-similarity (see e.g., [Cont, 2005](#) and [Mandelbrot, 2013](#)), we compare the MSM parameters estimated at the five-minute frequency with returns captured at ten-minute and fifteen-minute intervals. Table [A1](#) presents the AML parameter estimates of the MSM model with five-minute, ten-minute, and fifteen-minute frequencies¹⁶.

For each cryptocurrency, the number of volatility states $\bar{k}$ remains consistent across the different frequencies. This finding suggests that key aspects of volatility dynamics are stable across various high frequency scales. This consistency supports the hypothesis of self-similarity in financial markets, indicating that patterns observed at high frequencies may indeed be reflective of those at lower frequencies.

4 Comparison with Benchmark Models

4.1 In-sample Performance Comparisons

In this section, we investigate our model’s ability to match moments of cryptocurrency returns. We compare the moment matching performance of our model with two benchmark GARCH-type alternatives. We consider an exponential general autoregressive conditional heteroskedastic (EGARCH, [Nelson, 1991](#)) model and a Glosten-Jagannathan-Runkle

¹⁶For parameter estimation at the ten-minute frequency we used $D = 144$ and for the fifteen-minute frequency we used $D = 96$ in equation [\(3.2\)](#).

GARCH (hereafter GJR, [Glosten et al., 1993](#)) model. Both specifications furnish a well-balanced comparison for MSM. Similar to our baseline MSM model, both GJR and EGARCH generate autocorrelation in volatility of returns and have the same number of parameters as the MSM model. In addition, both GARCH models are easy to estimate by maximum likelihood (hereafter ML).

The benchmark GJR and EGARCH models both assume that cryptocurrency returns follow (3.2) and (3.1):

$$r_t = \mu_t + \sigma_t \varepsilon_t \quad (4.1)$$

$$\mu_t = \mu + \nu \overline{r_{(t-D):(t-1)}} + \rho r_{t-1} \quad (4.2)$$

for $t > D$ with $D = 288$ and $\mu_t = \mu/(1 - \nu - \rho)$ for $t \leq D$. The error terms $\{\varepsilon_t\}_{t=1}^T$ are independent and identically distributed Student-t variables with τ degrees of freedom. The GJR volatility specification is:

$$\sigma_t^2 = \omega + (\alpha + \gamma 1_{\{\varepsilon_{t-1} < 0\}}) \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2, \quad (4.3)$$

while the EGARCH volatility satisfies:

$$\log \sigma_t^2 = \omega + \alpha(|\varepsilon_{t-1}| + \gamma \varepsilon_{t-1}) + \beta \log \sigma_{t-1}^2. \quad (4.4)$$

Table 4 reports the ML parameter estimates of the extended GJR model in the in-sample period. Almost all parameters are significant at the 5% level, at the exception of μ (not significant for Dash and Ethereum) and ν (not significant for Bitcoin, Dash, and Ripple). Table 5 reports the ML parameter estimates of the extended EGARCH model in the in-sample period. All parameters are significant at the 5% level.

Table 4
Parameter Estimates of the GJR model via ML

	$\mu \times 10^4$	$\nu \times 10$	$\rho \times 10$	$\omega \times 10^8$	$\alpha \times 10$	β	$\gamma \times 10$	τ
BTC	0.075 (0.007)	-0.205 (0.115)	1.291 (0.013)	0.101 (0.005)	1.124 (0.012)	0.888 (0.001)	0.483 (0.016)	4.128 (0.021)
DASH	0.008 (0.020)	-0.028 (0.148)	0.259 (0.013)	7.583 (0.142)	1.348 (0.016)	0.865 (0.001)	0.331 (0.020)	3.580 (0.019)
ETH	0.029 (0.016)	0.606 (0.136)	1.227 (0.013)	4.827 (0.088)	1.384 (0.016)	0.860 (0.001)	0.275 (0.020)	3.987 (0.022)
LTC	-0.054 (0.018)	-0.446 (0.154)	0.349 (0.014)	3.794 (0.082)	1.222 (0.013)	0.878 (0.001)	0.268 (0.017)	4.115 (0.023)
XRP	-0.064 (0.016)	-0.021 (0.161)	0.197 (0.013)	1.605 (0.043)	1.175 (0.012)	0.882 (0.001)	0.400 (0.018)	4.023 (0.021)

This table presents parameter estimates of the GJR model via ML using five-minute cryptocurrency returns from September 1, 2016, 00:04 GMT to August 31, 2021, 23:59 GMT. Asymptotic standard errors are reported in parentheses.

Table 5
Parameter Estimates of the EGARCH model via ML

	$\mu \times 10^4$	$\nu \times 10$	$\rho \times 10$	ω	$\alpha \times 10$	β	$\gamma \times 10$	τ
BTC	0.199 (0.006)	-1.749 (0.083)	1.419 (0.012)	-0.217 (0.002)	1.889 (0.011)	0.994 (0.000)	-1.014 (0.038)	3.916 (0.021)
DASH	0.439 (0.013)	-1.148 (0.145)	0.272 (0.013)	-0.360 (0.003)	2.306 (0.015)	0.983 (0.000)	-0.482 (0.043)	3.475 (0.018)
ETH	-0.251 (0.012)	2.821 (0.109)	1.203 (0.013)	-0.374 (0.003)	2.340 (0.014)	0.983 (0.000)	-0.871 (0.040)	3.949 (0.021)
LTC	0.303 (0.012)	-2.870 (0.137)	0.344 (0.013)	-0.324 (0.003)	2.112 (0.013)	0.986 (0.000)	-0.644 (0.041)	4.024 (0.022)
XRP	-0.446 (0.010)	-4.995 (0.121)	0.172 (0.013)	-0.268 (0.002)	1.913 (0.011)	0.989 (0.000)	-0.428 (0.039)	3.837 (0.020)

This table presents parameter estimates of the EGARCH model via ML using five-minute cryptocurrency returns from September 1, 2016, 00:04 GMT to August 31, 2021, 23:59 GMT. Asymptotic standard errors are reported in parentheses.

Table 6
Comparison Across Models - In Sample

		Mean $\times 10^4$	Mom _D	Mom	S.D.	Exc.Kurt.	AC ₁	AC ₁ - AC ₂	H.-L.
BTC	MSM	0.145	0.011	0.119	0.002	1.775	0.422	0.035	112
	EGARCH	-0.520	-0.005	0.140	0.445	2.451	0.484	0.020	52
	GJR	0.408	-0.008	0.133	0.075	6.530	0.605	0.007	66.5
	Emp.	0.106	0.013	0.104	0.002	1.748	0.462	0.092	86
DASH	MSM	0.058	0.002	0.027	0.004	1.573	0.338	0.021	77
	EGARCH	0.641	-0.011	0.040	0.419	2.449	0.483	0.036	9
	GJR	0.231	-0.003	0.019	0.074	4.920	0.588	0.007	41
	Emp.	0.114	0.007	0.072	0.003	1.593	0.412	0.062	40
ETH	MSM	0.088	0.010	0.118	0.003	1.461	0.398	0.034	77
	EGARCH	-0.428	0.031	0.125	0.046	1.846	0.454	0.027	19
	GJR	0.060	0.009	0.122	0.012	3.216	0.586	0.019	37
	Emp.	0.147	0.016	0.132	0.003	1.590	0.442	0.082	28
LTC	MSM	-0.028	-0.003	0.042	0.003	1.440	0.371	0.024	114
	EGARCH	0.191	-0.014	0.034	0.046	1.707	0.409	0.014	28.5
	GJR	6.209	-0.014	0.041	282.623	7.403	0.611	0.004	78
	Emp.	0.120	0.013	0.119	0.003	1.523	0.411	0.050	52
XRP	MSM	-0.055	-0.002	0.021	0.004	1.614	0.417	0.024	141
	EGARCH	-0.425	-0.024	0.016	0.091	1.860	0.417	0.012	34
	GJR	8.675	-0.011	0.027	8.235	7.305	0.610	0.003	70.5
	Emp.	0.186	0.012	0.051	0.004	2.109	0.463	0.034	185

This table presents the medians of 100 simulated summary statistics produced by models and the empirical in-sample summary statistics for each cryptocurrency. For each statistic and cryptocurrency, we use bold characters to indicate the number that is closest to the empirical value.

To assess the accuracy of each model, we simulate 100 data series of size T from each estimated model reported in Tables 3-5, and for each simulated series, we calculate the eight summary statistics reported in Figure 4. We report the medians of the 100 simulated summary statistics in Table 6 and the empirical in-sample summary statistics for each cryptocurrency. For each statistic and cryptocurrency, we use bold characters to indicate the number that is closest to the empirical value. MSM provides better forecasts than EGARCH and GJR for the mean, daily momentum, standard deviation, excess kurtosis, and speed of decay in autocorrelations of absolute returns ($AC_1 - AC_2$). Momentum is relatively well predicted by all models. First autocorrelations in volatility (AC_1) and half-lives AC ($H. - L.$) do not seem to clearly show MSM's superiority. To better understand the volatility persistence behaviour, we plot in Figure A1 the autocorrelations in absolute returns for up to 100 lags in both simulated and empirical series. Figure A1 shows that MSM is the best at capturing volatility persistence. Overall, MSM is the best among the three models at capturing cryptocurrency return dynamics.

4.2 Out-of Sample Forecast Comparisons

In this section, we evaluate the out-of-sample performance of our model and compare it with the benchmarks outlined in 4.1. Using the previously obtained in-sample parameter estimates, we evaluate the prediction powers of the MSM, EGARCH and GJR models in the out-of-sample period $t = T + 1, \dots, T'$. For the GARCH-type specifications, on each out-of-sample date t , we generate $N = 10^5$ forecasts $\{\tilde{r}_t^{(n)}\}_{n=1}^N$ from (4.1)-(4.4) using the parameter estimates obtained in the previous subsection, and past returns $r_{1:(t-1)}$.

To forecast the return at date t using the MSM model, we first need to simulate the M_t states. We use a particle filter and AML parameter estimates obtained in Section 3.3 and on each date t generate $N = 10^5$ simulated $\{\tilde{M}_t^{(n)}\}_{n=1}^N$ and $\{\tilde{r}_t^{(n)}\}_{n=1}^N$ to finitely approximate the distribution of r_t conditional on knowing past observations $r_{1:(t-1)}$. The particle filtering algorithm is defined in Appendix A.1.

We start by examining the accuracy of the predicted conditional distributions of the returns. For each model and each date $t = T+1, \dots, T'$, we construct 90% prediction intervals:

$$PI_t(90\%) = \left[q_{0.05}(\tilde{r}_t^{(1)}, \dots, \tilde{r}_t^{(N)}), q_{0.95}(\tilde{r}_t^{(1)}, \dots, \tilde{r}_t^{(N)}) \right],$$

where $q_\alpha(\cdot)$ indicates the α -th sample quantile. We then calculate the 90%-coverage probabilities, that is, the proportion of times that the actual returns are in the 90%-prediction intervals:

$$\text{COV}(90\%) = \frac{1}{T' - T} \sum_{t=T+1}^{T'} 1_{\{r_t \in PI_t(90\%)\}}.$$

The first column of Table 7 reports the 90%-coverage probabilities. MSM systematically provides better prediction intervals than its GARCH counterparts, i.e. it provides 90%-coverage probabilities that are the closest to 90%.

Next, we evaluate the performance of the value-at-risk forecasts. At each date $t = T + 1, \dots, T'$, we calculate the value-at-risk (VaR) forecasts defined as

$$\text{VaR}_{\alpha,t} = -q_\alpha(\tilde{r}_t^{(1)}, \dots, \tilde{r}_t^{(N)}),$$

with $\alpha = 1\%$, $\alpha = 5\%$ and $\alpha = 10\%$. We report in columns 2 to 4 of Table 7 the failure rates of the $\text{VaR}_{\alpha,t}$:

$$p_\alpha = \frac{1}{T' - T} \sum_{t=T+1}^{T'} 1_{\{r_t < (-\text{VaR}_{\alpha,t})\}}.$$

The closer p_α is to α , the better the forecasts. MSM outperforms the VaR forecasts provided by EGARCH and GJR, i.e. MSM provides p_α values that are the closest to their theoretical α values.

Further, we investigate the accuracy of the volatility forecasts. We provide results of Mincer-Zarnowitz regressions of $|r_t^d| = \delta + \eta \mathbb{E}_{t-1}(|r_t^d|) + e_t$ where r_t^d is the demeaned return at date t . The forecasted means $\mathbb{E}_{t-1}(\cdot)$ given past observations $r_{1:(t-1)}$ are estimated with

sample means obtained with the forecasted $\{\hat{r}_t^{(n)}\}_{n=1}^N$. For unbiased forecasts, we expect to have $\delta = 0$ and $\eta = 1$. Columns 5 and 6 of Table 7 report the regression intercepts and slopes. The last column reports the pseudo- R^2 value (in %) under the restrictions $\delta = 0$ and $\eta = 1$ defined as:

$$\text{Pseudo-}R^2 = 1 - \frac{SSE}{SSR},$$

where SSE is the sum of squared forecast errors and SSR is the sample variance of the absolute values of the demeaned returns multiplied by $(T' - T - 1)$. These pseudo- R^2 values can turn negative when forecasts are very inaccurate. In almost all instances, MSM performs best. Most MSM δ values are closer to zero, all η values are closer to one and all pseudo- R^2 values are closer to 100% than their GARCH-type alternatives, often by large margins.

Table 7
Forecast Comparison Across Models

Model	90%-Coverage probabilities	Failure rates of VaR forecasts (in %)			Volatility forecasts		
		1%	5%	10%	$\delta \times 1000$	η	Pseudo- R^2 (in %)
BTC	MSM	91.23	4.48	9.50	0.0074	0.9746	20.68
	GJR	93.85	3.10	6.83	0.2160	0.5900	5.49
	EGARCH	96.30	1.87	4.58	0.1987	0.5323	-7.16
DASH	MSM	91.92	4.18	9.13	0.0101	0.9402	20.49
	GJR	97.00	1.53	4.06	0.1412	0.5749	-8.38
	EGARCH	97.47	1.34	3.72	0.5174	0.3794	-44.05
ETH	MSM	91.63	4.25	8.73	0.0249	0.9311	21.25
	GJR	95.72	2.15	4.99	0.1139	0.6504	6.64
	EGARCH	96.57	1.67	3.94	0.2034	0.5474	-6.39
LTC	MSM	90.99	4.43	9.20	-0.0359	1.0180	21.39
	GJR	96.08	1.93	4.45	0.2028	0.5707	-1.27
	EGARCH	96.51	1.75	4.23	1.1950	0.0087	-3680.80
XRP	MSM	91.18	4.57	9.10	-0.0453	1.0040	23.08
	GJR	95.97	2.10	4.94	0.2288	0.5563	-1.83
	EGARCH	96.87	1.61	3.91	0.0052	0.6596	3.61

The table presents the out-of-sample forecasting performance of the MSM, GJR, and EGARCH models. Column 1 presents the coverage probabilities of 90% prediction intervals for the out-of-sample five-minute returns. Columns 2 to 4 present the failure rates of 1%, 5% and 10% value-at-risk predictions. Column 5 and Column 6 correspond to OLS parameter estimates of the Mincer-Zarnowitz regression $|r_t^d| = \delta + \eta \mathbb{E}_{t-1}(|r_t^d|) + e_t$ where r_t^d is the demeaned return at date t . For an unbiased result, we expect to have $\delta = 0$ and $\eta = 1$. Column 7 reports pseudo- R^2 (in %) under the restrictions $\delta = 0$ and $\eta = 1$.

5 Trading Strategy

This section first introduces a trading strategy. The trading signal at date t , where t indexes five-minute intervals, is defined as the ratio of the forecasted expectation to the forecasted volatility of the return for date $t + 1$. Next, we evaluate the performance of our trading strategy when the forecasted returns are generated from the three models in Section 3 as well as from a constant volatility model. We test the robustness of our trading strategy under various scenarios including cryptocurrency crash days and execution delays.

5.1 Trading Signal Generation

To simplify the presentation, we first consider trading one single cryptocurrency. Our trading period starts on the last in-sample date T (23:59 GMT, August 31, 2021) and ends on the last out-of-sample date T' (23:59 GMT, August 31, 2023). At each date $t = T, \dots, T'$, our entire capital is invested in either a long or a short spot position. We denote the position type at date $t > T$ (ordered at date $t - 1$) as follows:

$$O_t = \begin{cases} +1 & \text{long position} \\ -1 & \text{short position} \end{cases}.$$

On each date $t > T$, we compute the *dynamic Sharpe ratio trading signal* for date $t + 1$:

$$S_{t+1} = \frac{\mathbb{E}_t(r_{t+1})}{\sigma_t(r_{t+1})},$$

where $\mathbb{E}_t(r_{t+1})$ and $\sigma_t(r_{t+1})$ are estimated with the sample mean and standard deviation of $\{\tilde{r}_{t+1}^{(n)}\}_{n=1}^N$ generated in Section 4.2 using the associated MSM, EGARCH, or GJR models. A large positive value of S_{t+1} indicates a good opportunity to buy, while a large negative value of S_{t+1} , a good opportunity to short sell. We therefore check whether the absolute value of

the Sharpe ratio $|S_{t+1}|$ is larger than the threshold $|S|^{1/D}$, defined as the top $1/D$ -quantile of the absolute values of the Sharpe ratios calculated in the in-sample period with $D = 288$. This choice of D is motivated by the fact that, on average, $|S|^{1/D}$ is exceeded once per day. If the trading signal satisfies the condition $|S_{t+1}| > |S|^{1/D}$, we trade according to the sign of the predicted return. Conversely, if this condition is not met, we maintain the current position without making any adjustments. Concretely, our trading strategy is defined by the following iterative procedure.

Dynamic Sharpe Ratio (DSR) Trading Strategy

At date T : open a position for date $T + 1$: $O_{T+1} = \text{sign}[\mathbb{E}_T(r_{T+1})]$.

$$\text{At date } t > T: \text{ order the position for } t + 1: O_{t+1} = \begin{cases} O_t & \text{if } |S_{t+1}| \leq |S|^{1/D} \\ \text{sign}(S_{t+1}) & \text{if } |S_{t+1}| > |S|^{1/D} \end{cases} .$$

The DSR trading can be applied to any model that dynamically captures the mean of the high frequency returns. A dynamic mean μ_t is necessary as otherwise if the mean $\mu_t = \mu$ is constant, the Sharpe ratio signal S_{t+1} never changes its sign and the DSR strategy reduces to a passive trading.¹⁷

In order to evaluate the profitability of the DSR trading strategy under various models, we compute cumulative returns under a predetermined trading fee structure that captures both explicit transaction costs and implicit costs such as the bid-ask spread. We consider two types of trading fees: an opening fee $\tau^O \in [0, 1]$, which incorporates the bid-ask spread and is paid each time a position is opened, and an hourly rollover fee $\tau^R \in [0, 1]$, which is incurred at the end of every hour that a position remains open. Both fees are expressed as

¹⁷A volatility model assuming that the high frequency return has a constant mean ($\mathbb{E}_t(r_t) = \mu$ for all t) or even zero mean such as the heterogeneous autoregressive (HAR) model of [Corsi \(2009\)](#) will never allow the DSR to switch between long and short positions.

a percentage of the invested capital.

Let L be the total number of switches in positions during the trading period, and let $\{d_j\}$ be the execution dates of the position switches with $T = d_1 < \dots < d_L < T'$. By default, $d_1 = T$ corresponds to the first trading order date. For $j = 2, \dots, L$, the switches satisfy $O_{d_j} O_{d_{j+1}} < 0$. Furthermore, by convenience, we denote $d_{L+1} = T'$ the last date in the out-of-sample period.

Let us further denote R_t^j the cumulative return on the cryptocurrency over the period $[d_j, t]$ with $d_j < t \leq d_{j+1}$, corresponding to a long position placed at date d_j with an hourly rollover fee $\tau^R \in [0, 1]$. The gross return B_t^j is defined as:

$$B_t^j = 1 + R_t^j = \prod_{t'=d_j+1}^t (1 + r_{t'}) (1 - \tau^R 1_{\{t' \equiv d_j \pmod{12}\}}), \quad (5.1)$$

where the indicator function $1_{\{t' \equiv d_j \pmod{12}\}}$ takes the value one every time 12 five-minute intervals have elapsed, corresponding to an hourly rollover fee payment, and zero otherwise.

Suppose that the holding period $[d_j, d_{j+1}]$ corresponds to a short position. We assume that the cumulative return of a short position over the period $[d_j, t]$ for $d_j < t \leq d_{j+1}$ is the negative of the cumulative return of a long position. Then, the gross return on the short position over the period $[d_j, t]$ is:

$$D_t^j = 1 - R_t^j = 2 - B_t^j. \quad (5.2)$$

We track the cumulative return on a period $[T, t]$ at any given date $t = T, \dots, T'$. Opening fees $\tau^O \in [0, 1]$ are paid each time we open a new position. At the first trading date $t = T$, the capital is set to $C_T = 1$ (USD). The capital C_t at date $t \geq T + 1$ under the Dynamic Sharpe Ratio (DSR) trading defined above can be expressed in the following recursive way. For a given t , let j be such that $d_j < t \leq d_{j+1}$, $j = 1, \dots, L$. The capital C_t is given by:

$$C_t = \begin{cases} (1 - \tau^O) C_{d_j} B_t^j & \text{if } O_t = 1 \\ (1 - \tau^O) C_{d_j} D_t^j & \text{if } O_t = -1 \end{cases}. \quad (5.3)$$

For our first illustration, we set the trading fees to $\tau^O = 0.05\%$ and $\tau^R = 0.00125\%$, reflecting typical fees on major cryptocurrency platforms such as Binance and Kraken.¹⁸ For each cryptocurrency, we calculate the cumulative return $r_{cum,T'} = C_{T'} - 1$ for the two-year out-of-sample period from September 1, 2021 to August 31, 2023 using our three models. As a market benchmark, we also report the buy-and-hold strategy for each cryptocurrency. In addition to the cumulative returns, we also report the annualized static Sharpe ratio with zero risk-free rate:

$$S_{sta} = \frac{\bar{r}}{s} \sqrt{288 \cdot 365},$$

where

$$\bar{r} = \frac{1}{T' - T} \sum_{t=T}^{T'-1} \frac{C_{t+1}}{C_t} - 1 \quad \text{and} \quad s = \sqrt{\frac{1}{T' - T - 1} \sum_{t=T}^{T'-1} \left(\frac{C_{t+1}}{C_t} - 1 - \bar{r} \right)^2}.$$

¹⁸Fee breakdown can be found at <https://support.kraken.com/hc/en-us/articles/206161568> for Kraken, and at <https://www.binance.com/en/fee/marginFee> for Binance. These platforms are used in the current literature to test well-established theories. For example, [Ardia et al. \(2024\)](#) contribute by applying bid-ask spread estimators to high frequency cryptocurrency pairs traded on Binance.

Table 8
Performances of DSR and Passive Trading Strategies

		BTC	DASH	ETH	LTC	XRP
DSR-MSM	r_{cum}	681.98	1199.85	390.93	1225.80	524.27
	S_{sta}	2.33	2.20	1.63	2.25	1.68
DSR-GJR	r_{cum}	-29.10	-18.13	177.37	328.79	291.09
	S_{sta}	-0.09	0.21	1.15	1.40	1.33
DSR-EGARCH	r_{cum}	173.91	-88.67	227.42	-76.65	26.09
	S_{sta}	1.26	-1.15	1.30	-0.68	0.49
DSR-ConstVol	r_{cum}	124.07	-39.98	256.06	173.27	-21.96
	S_{sta}	1.06	0.05	1.36	1.07	0.19
Passive	r_{cum}	-44.65	-88.36	-51.69	-62.99	-57.05
	S_{sta}	-0.40	-1.12	-0.29	-0.36	-0.23

This table presents two metrics to quantify the portfolio performance: the cumulative return r_{cum} (in %) and the annualized static Sharpe ratio S_{sta} . These metrics are computed for the out-of-sample period from September 1, 2021, 00:04 GMT to August 31, 2023, 23:59 GMT. The fee of opening a position is 0.05% and the hourly fee of maintaining a position is 0.00125%. These fees are not applied to the buy-and-hold strategy.

Table 8 reports the out-of-sample performance of our DSR trading strategy under the MSM, GJR, and EGARCH models. To assess the importance of accurately capturing the volatility when trading cryptocurrencies, we also report in Table 8 the DSR trading strategy under a Constant Volatility (ConstVol) model.¹⁹ Table 8 also reports out-of-sample performance of the buy-and-hold strategy for the five cryptocurrencies. The DSR trading approach incorporating the MSM (DSR-MSM) yields the highest cumulative returns and static Sharpe ratios for all cryptocurrencies. Notably, the annualized static Sharpe ratio of the DSR-MSM

¹⁹The Constant Volatility model assumes that returns are generated under $r_t = \mu_t + \sigma u_t$ with μ_t defined in equation (3.2) and $\{u_t\}$ are independent standard normal variables. Parameters in (3.2) and σ are estimated by OLS.

trading strategy is greater than 1 for all cryptocurrencies.

5.2 Return Decomposition: Short vs Long

We further examine the dynamics of returns by decomposing the overall return into two parts: returns derived from long positions and those from short positions. This decomposition ensures the resilience of our trading benchmarks across various market conditions such as bear market throughout 2022 and bull market in early 2023.

Table 9
Return Decomposition of the DSR Trading: Short and Long

		BTC	DASH	ETH	LTC	XRP
DSR-MSM	Long	133.26	80.60	98.50	206.43	131.46
	Short	235.24	619.73	147.32	332.66	169.71
DSR-GJR	Long	-41.95	-66.42	29.47	79.55	64.00
	Short	22.13	142.78	114.23	138.82	138.46
DSR-EGARCH	Long	28.37	-89.59	55.53	-72.56	0
	Short	113.37	8.83	110.52	-14.91	26.09
DSR-ConstVol	Long	29.78	-43.22	53.05	46.09	-24.13
	Short	72.66	5.72	132.64	87.06	2.87

This table presents the cumulative returns on long and on short positions in percentage, for each cryptocurrency, when using the DSR trading with MSM, GJR, EGARCH, and ConstVol models. The cumulative returns are computed for the out-of-sample period from September 1, 2021, 00:04 GMT to August 31, 2023, 23:59 GMT. The fee of opening a position is 0.05% and the hourly fee of maintaining a position is 0.00125%.

Table 9 reports the cumulative returns on long positions ($C_{\text{long},T'} - 1$) and on short positions ($C_{\text{short},T'} - 1$) in percentage (see Appendix A.2 for calculation details), for each cryptocurrency, when using the DSR trading with MSM, GJR, EGARCH, and ConstVol models. This table shows that cumulative returns obtained from long and short positions

are closely aligned with each other across cryptocurrencies under the DSR trading with MSM. This result highlights the adaptability of the DSR-MSM trading to various market conditions as it is able to capture returns from both upward and downward market movements.

5.3 Return Decomposition: Crash vs Stable Days

Since cryptocurrencies are known for their extreme negative price swings, often referred to as *crashes*, understanding how trading strategies perform in these tail-event scenarios can offer insight into their robustness and risk profiles. In particular, [Biais et al. \(2023\)](#) highlight the importance of modeling price crashes in crypto markets for both theoretical and practical considerations. However, rather than relying on exogenous event-based definitions of extreme moves, we follow a purely return-based approach to identify days on which a cryptocurrency experiences large negative returns.

Formally, for each cryptocurrency, we first compute its daily returns over the out-of-sample period from September 1, 2021, 00:04 GMT to August 31, 2023, 23:59 GMT by aggregating the five-minute price changes into daily returns. We define day i as a crash-day if $r_{day,i} \leq -10\%$ and stable-day otherwise, where $r_{day,i}$ denotes the daily return on day i . Over the 730 out-of-sample days, we detect 4 BTC, 21 DASH, 10 ETH, 13 LTC, and 13 XRP crash-days. To compare the performances of our DSR trading strategies under our model and benchmark models, we compute the daily geometric mean returns of the strategies on crash days and stable days separately.²⁰

²⁰Specifically, if $r_{day,1}, r_{day,2}, \dots, r_{day,K}$ are the strategy's daily returns in a subset of K days (e.g., all crash-days), we calculate the geometric mean return as:

$$r_{\text{geo}} = \left(\prod_{i=1}^K (1 + r_{day,i}) \right)^{\frac{1}{K}} - 1$$

Table 10

Geometric Mean Returns on Crash and Stable Days

		BTC	DASH	ETH	LTC	XRP
DSR-MSM	Crash	4.16	6.97	6.83	5.66	4.10
	Stable	0.26	0.16	0.13	0.26	0.18
DSR-GJR	Crash	-8.13	-6.30	1.54	5.61	-3.60
	Stable	-0.00	0.16	0.12	0.10	0.26
DSR-EGARCH	Crash	0.66	-11.57	2.67	-12.25	6.93
	Stable	0.14	0.06	0.13	0.04	-0.09
DSR-ConstVol	Crash	8.04	3.29	1.52	1.56	-1.60
	Stable	0.07	-0.17	0.16	0.11	-0.01

This table presents the geometric means of daily returns during crash days and stable days in percentage, for each cryptocurrency, when using the DSR trading with MSM, GJR, EGARCH, and ConstVol models. The geometric means are computed for the out-of-sample period from September 1, 2021, 00:04 GMT to August 31, 2023, 23:59 GMT. The fee of opening a position is 0.05% and the hourly fee of maintaining a position is 0.00125%.

Table 10 presents the resulting geometric mean daily returns of the DSR trading strategy using MSM, GJR, EGARCH, and ConstVol models. The table shows that DSR-MSM strategy is the safest strategy to adopt as it is the only one to provide positive gains under both crash and stable days for all five cryptocurrencies. In addition, this table shows that crash days are an opportunity for DSR-MSM to gain even more than on stable days, underscoring the importance of capturing multi-scale volatility features, particularly in the cryptocurrency market prone to abrupt downward moves.

5.4 Portfolio Trading

Having established our trading framework for individual cryptocurrencies, we now extend the strategy to a multi-asset setting. In this section, we apply the DSR trading strategy to a diversified portfolio, allocating capital across multiple cryptocurrencies to assess the overall performance. Appendix A.3 explains the calculation of cumulative portfolio returns of the DSR trading strategy and corresponding annualized static Sharpe ratios.

Figure 6 presents the cumulative portfolio returns of the DSR trading strategy with four models with trading fees of $\tau^O = 0.05\%$ and $\tau^R = 0.00125\%$ between September 1, 2021, 00:04 GMT to August 31, 2023, 23:59 GMT. The buy-and-hold strategy produces a cumulative return of -61% , since the cryptocurrency market did experience a severe downturn over the out-of-sample period, see Figure 1. The DSR-MSM, DSR-GJR, DSR-EGARCH, and DSR-ConstVol approaches produce, respectively cumulative portfolio returns of 805% , 150% , 52% , and 84% with annualized static Sharpe ratios of 3.05 , 1.31 , 0.81 , and 0.99 .

The dominance of DSR-MSM over the other strategies is not limited to the trading fees that we used to perform our previous analysis. We further investigate the robustness of the DSR-MSM trading strategy under different rollover fees. Figure 7 illustrates the cumulative return (left panel) and the annualized static Sharpe ratio (right panel) of the DSR trading strategy for various rollover fees τ^R , under fixed opening fee $\tau^O = 0.05\%$. The DSR trading strategy with MSM outperforms the other three strategies for all rollover fees, and provides positive portfolio returns for hourly rollover fees $\tau^R < 0.0138\%$, which is significantly higher than fees charged by major cryptocurrency exchanges. In addition, DSR-MSM is uniformly profitable across time as the annualized static Sharpe ratio is greater than 1 for hourly rollover fees $\tau^R < 0.0103\%$, while this is only the case for $\tau^R < 0.0027\%$ using GJR, $\tau^R < 0.0006\%$ using EGARCH, and $\tau^R < 0.0012\%$ using constant volatility.

In order to streamline our discussion, we now concentrate solely on the DSR-MSM approach. Up to this point, we have analyzed the profitability under various rollover fees while keeping the opening fee at $\tau^O = 0.05\%$. Figure 8 illustrates the DSR-MSM's performance

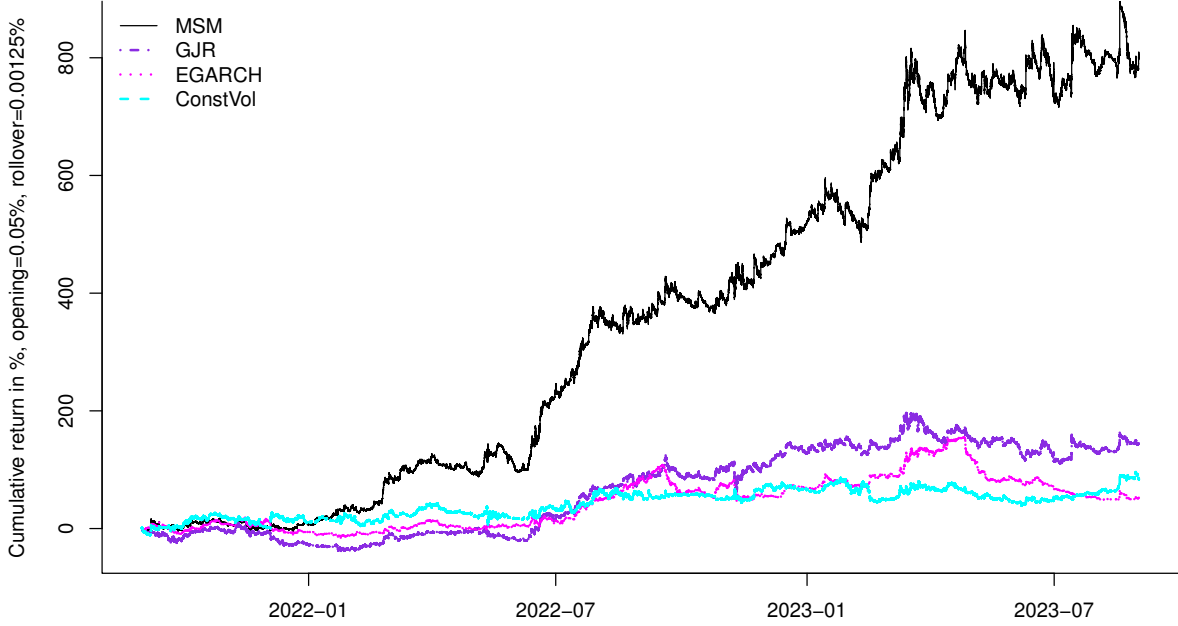


Fig. 6. *Cumulative Portfolio Returns.* Notes: This figure illustrates the cumulative returns of the DSR trading with MSM (continuous line), GJR (dash-dotted line), EGARCH (dotted line), and ConstVol (dashed line) models. Trading runs through the period from September 1, 2021 to August 31, 2023. The y-axis represents cumulative returns in percentage.

for opening fees of $\tau^0 = 0.05\%, 0.1\%, 0.2\%$. As expected, cumulative returns (left panel) and annualized static Sharpe ratios (right panel) decline with higher opening fees. Nevertheless, DSR-MSM produces positive returns even when the opening fee is as large as $\tau = 0.2\%$ and the hourly rollover fee is $\tau^R < 0.011\%$. In addition, DSR-MSM trading provides annualized static Sharpe ratios that are greater than 1 even if the opening fee is as large as $\tau^0 = 0.2\%$ and the hourly rollover fee is $\tau^R < 0.0072\%$.

5.5 Trading with Execution Delays

Until now, our analysis assumed instantaneous trading executions. In reality, however, an order often experiences a delay before it is fully executed, which may affect the performance of high-frequency trading strategies. To account for this, in this section, we extend our framework by incorporating a fixed execution delay. To incorporate execution delays into the single-asset capital equation formula C_t introduced in Section 5.1, we first assume that

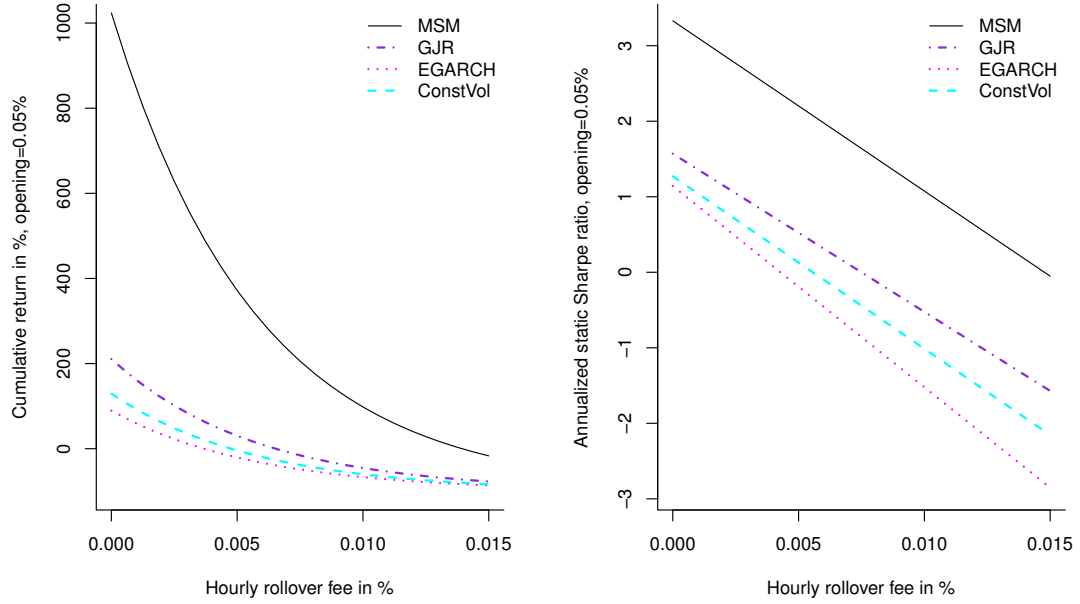


Fig. 7. *Portfolio Performance Under Various Rollover Fees.* Notes: This figure presents the cumulative returns (left panel) and annualized static Sharpe ratios (right panel) of the DSR trading with MSM (continuous lines), GJR (dash-dotted lines), EGARCH (dotted lines), and ConstVol (dashed line) models for various hourly rollover fees. Trading period is from September 1, 2021, 00:04 GMT to August 31, 2023, 23:59 GMT.

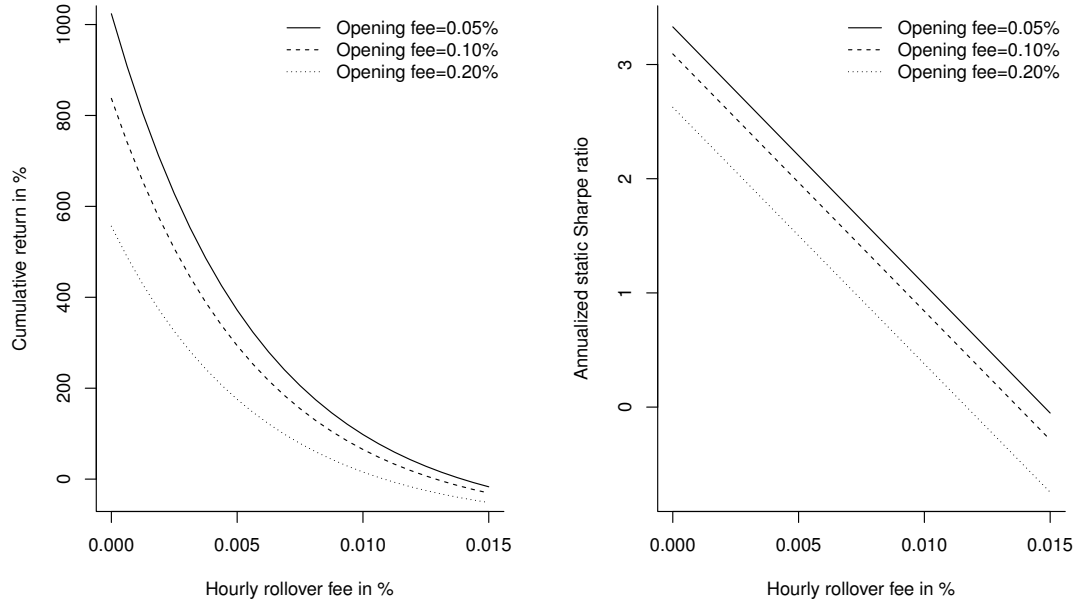


Fig. 8. *DSR-MSM Trading Under Various Trading Fees.* Notes: This figure presents the cumulative returns (left panel) and annualized static Sharpe ratios (right panel) of the DSR-MSM trading for various opening fees $\tau^0 = 0.05\%$ (continuous lines), $\tau^0 = 0.1\%$ (dashed lines), and $\tau^0 = 0.2\%$ (dotted lines). Trading period is from September 1, 2021, 00:04 GMT to August 31, 2023, 23:59 GMT.

the first opening execution is delayed by $\Delta \times 5$ minutes from the decision date $t = T$. At each switching execution date, the position closing is delayed by $\Delta \times 5$ minutes and the opening of the new position by $2\Delta \times 5$ minutes from the decision date t , $\Delta \in [0, 0.5]$. Since prices on CoinMarketCap are available only at intervals of five minutes or longer, we assume that cryptocurrency prices move linearly within each five-minute interval. See Appendix A.4 for detailed derivations and adjustments to the capital equation C_t introduced in Section 5.1 under the execution delays.

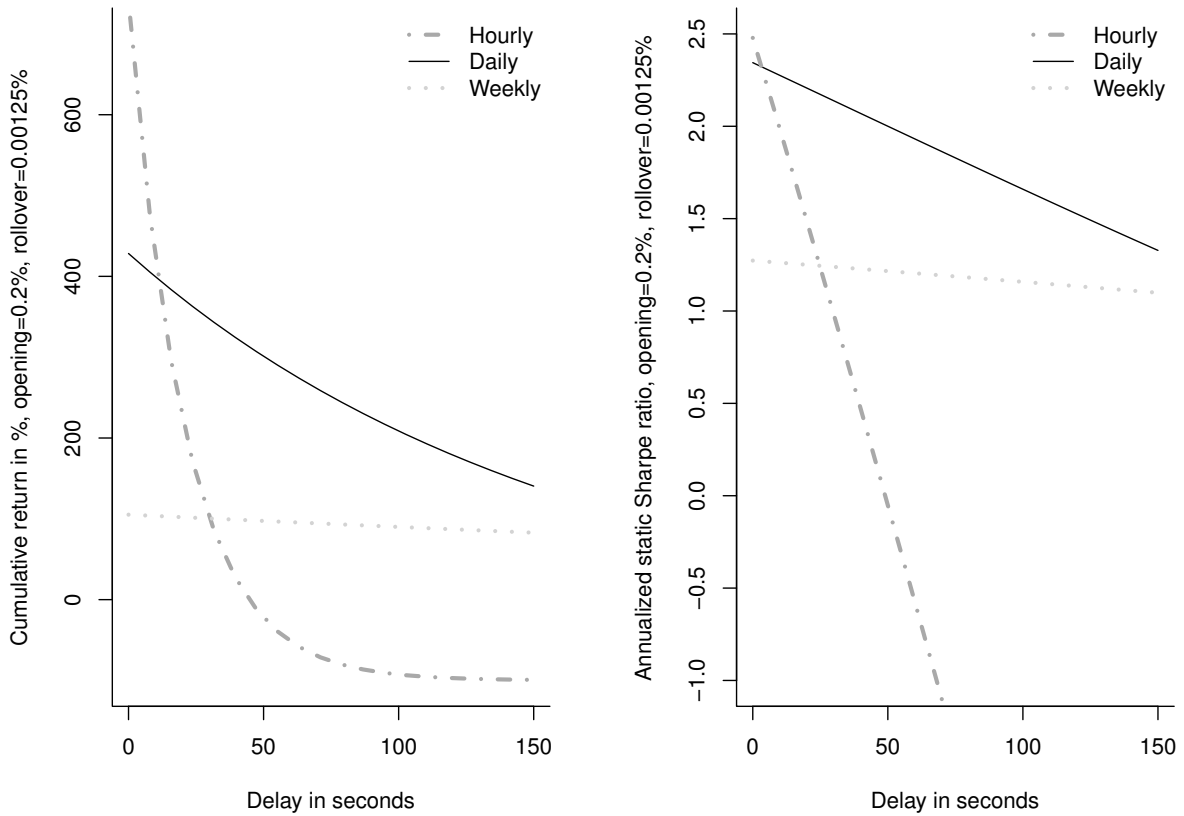


Fig. 9. *Trading with Execution Delays.* Notes: Cumulative returns (left panel) and annualized static Sharpe ratios (right panel) for DSR-MSM are plotted for various execution delays between 0 and 150 seconds. Daily trading ($D = 288$, continuous lines) is benchmarked with hourly ($D = 12$, dashed-dotted lines) and weekly ($D = 2,016$, dotted lines) tradings. Trading takes place over the period from September 1, 2021, 00:04 GMT to August 31, 2023, 23:59 GMT. Opening fee is 0.2%, and rollover fee is 0.00125%.

Figure 9 displays the cumulative portfolio return (left panel) and annualized static Sharpe ratio (right panel) of the DSR-MSM trading under various execution delays at the end of the two-year trading period from September 1, 2021, 00:04 GMT to August 31, 2023, 23:59

GMT. In addition to the daily trade frequency ($D = 288$), we also report the results for hourly ($D = 12$), and weekly ($D = 2,016$) trading frequencies in Figure 9. This figure shows that DSR-MSM at the daily frequency can withstand delays up to 150 seconds. Furthermore, DSR-MSM at the hourly frequency provides larger gains than daily trading if execution delays are less than 12 seconds, but these gains rapidly decay. On the other hand, not surprisingly, DSR-MSM at the weekly frequency is barely impacted by the execution delays less than 150 seconds, but gains are smaller than those provided by DSR-MSM at the daily frequency.

6 Concluding Remarks

We model high frequency cryptocurrency returns with a multifractal process. We use AML to estimate the model parameters. We investigate the link between multifractal model parameters and structural features of each cryptocurrency network. The volatility persistence is associated with the age of cryptocurrency coins and a higher concentration of wealthy holders. Estimates across different sampling frequencies remain largely consistent, underscoring the self-similar nature of cryptocurrency price dynamics. We show that cryptocurrency volatility often exhibits multiple overlapping persistence channels, which the multifractal model is able to capture more effectively. Out-of-sample tests confirm that the multifractal specification delivers more accurate forecasts of returns, coverage probabilities, and Value-at-Risk failure rates than conventional benchmarks. Finally, we introduce a Dynamic Sharpe Ratio (DSR) trading methodology, fed by model-implied forecasts of mean and volatility. When implemented with the multifractal model, DSR produces strong profitability under realistic market conditions, including trading fees and delayed executions. The gains persist even during periods of severe market downturns.

A natural question arises at this point. Can we apply multifractal models to other crypto-native assets such as Non-Fungible Tokens (NFTs) and utility tokens? (see, e.g., [Sockin and Xiong 2023](#), [Abadi and Brunnermeier 2024](#), and [Goldstein et al. 2024](#)) As of today, due to

limited secondary markets, the daily trading volumes of NFT collections and utility tokens are still insufficiently low to generate meaningful high frequency data points compared to cryptocurrencies with Layer 1 blockchain protocols, such as Bitcoin and Ethereum. Our approach can serve as a meaningful starting point for future studies when other crypto-asset markets mature.

Another avenue for future work lies in expanding the multifractal framework to capture more nuanced microstructure features in cryptocurrency markets. For example, decentralized exchanges (DEXs) have time-varying transaction fees, and block-level settlement can lead to nontrivial delays or congestion effects. Incorporating such dynamics would require modeling a heterogeneous distribution of execution speeds, where the effective cost and timing of trades respond to network conditions and liquidity frictions (see e.g., [Barbon and Ranaldo 2022](#) and [Hasbrouck et al. 2025](#)). The shifting transaction environments further shape high frequency volatility patterns and merit additional extensions of the multifractal approach.

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A Appendix

A.1 Bootstrap Particle Filter

Once the structural parameters are estimated by AML, we can use a bootstrap particle filter [Gordon et al. \(1993\)](#) to impute the hidden states and to forecast future returns with the MSM model. The algorithm is defined as follows.

At date $t = 0$, simulate N particles $\{M_0^{(n)}\}_{n=1}^N$ from the stationary distribution λ_0 provided in [Section 3.1](#). For $t = 1, \dots, T'$, recursively iterate the following three steps.

Step 1 (Propagation): For every $n = 1, \dots, N$, simulate a particle $\tilde{M}_t^{(n)}$ from $M_{t-1}^{(n)}$ using the transition probabilities provided in [Section 3.1](#).

Step 2 (Correction): Given the new data point r_t , compute for each $\tilde{M}_t^{(n)}$ the importance weight

$$w_t^{(n)} = \frac{1}{\sigma_t^{(n)} \sqrt{2\pi}} \exp \left\{ -\frac{1}{2} \left(\frac{r_t - \mu_t}{\sigma_t^{(n)}} \right)^2 \right\}, \quad n \in \{1, \dots, N\},$$

where μ_t is given in [\(3.2\)](#) and

$$(\sigma_t^{(n)})^2 = \sigma^2 \prod_{k=1}^{\bar{k}} \tilde{M}_{k,t}^{(n)}. \quad (\text{A.1})$$

Step 3 (Selection): For every $n = 1, \dots, N$, draw $M_t^{(n)}$ from $\tilde{M}_t^{(1)}, \dots, \tilde{M}_t^{(N)}$ with importance weights $p_t^{(1)}, \dots, p_t^{(N)}$, where $p_t^{(n)} = w_t^{(n)} / \sum_{n'=1}^N w_t^{(n')}$.

The particles $\{\tilde{M}_t^{(n)}\}$ generated in Step 1 are a finite approximation of the distribution of M_t given past observations $r_{1:(t-1)}$. The future return r_t can then be forecasted by drawing $\tilde{r}_t^{(n)}$ for each $n = 1, \dots, N$ from $\mathcal{N}(\mu_t, (\sigma_t^{(n)})^2)$, where $\sigma_t^{(n)}$ is given in [\(A.1\)](#). The set of forecasts $\{\tilde{r}_t^{(n)}\}_{n=1}^N$ finitely approximates the distribution of $r_t | r_{1:(t-1)}$.

A.2 Calculation of Return Decomposition: Short vs Long

We first set initial capitals for long positions to $C_{\text{long},T} = 1$ and for short positions to $C_{\text{short},T} =$

1. For a given t , let j be such that $d_j < t \leq d_{j+1}$, then the cumulative return on long positions only (i.e. the position type at date t is $O_{\text{long},t} = 1_{\{O_t=+1\}}$) is

$$C_{\text{long},t} = \begin{cases} (1 - \tau^0)C_{\text{long},t-1}B_t^j & \text{for } O_t = +1 \\ C_{\text{long},t-1} & \text{otherwise} \end{cases} \quad (\text{A.2})$$

where B_t^j is defined in (5.1). The cumulative return on short positions only (i.e. the position type at date t is $O_{\text{short},t} = -1_{\{O_t=-1\}}$) is

$$C_{\text{short},t} = \begin{cases} (1 - \tau^0)C_{\text{short},t-1}D_t^j & \text{for } O_t = -1 \\ C_{\text{short},t-1} & \text{otherwise} \end{cases} \quad (\text{A.3})$$

where D_t^j is defined in (5.2).

A.3 Calculation of Cumulative Portfolio Returns and Static Sharpe Ratios

At date T , we allocate the same proportion of our wealth to each of the five cryptocurrencies, $W_{i,T} = 0.2$, $i = 1, \dots, 5$ and for each cryptocurrency, we apply our DSR trading strategy described in the Section 5.1 using our three models. The portfolio capital at time $t = T + 1, \dots, T'$ evolves according to:

$$C_t^P = \sum_{i=1}^5 W_{i,T} C_{i,t},$$

where $C_{i,t}$ is calculated with formula (5.3) using cryptocurrency i , and the cumulative portfolio return at date t is

$$r_{cum,t}^P = C_t^P - 1.$$

The annualized static Sharpe ratio of the portfolio for the trading period $[T, T']$ is:

$$S_{sta}^P = \frac{\overline{r^P}}{s^P} \sqrt{288 \cdot 365},$$

where

$$\overline{r^P} = \frac{1}{T' - T} \sum_{t=T}^{T'-1} \frac{C_{t+1}^P}{C_t^P} - 1, \quad \text{and} \quad s^P = \sqrt{\frac{1}{T' - T - 1} \sum_{t=T}^{T'-1} \left(\frac{C_{t+1}^P}{C_t^P} - 1 - \overline{r^P} \right)^2}.$$

A.4 Calculation of Returns under Execution Delays

Following the linearity assumption, in-between two consecutive dates the price of a cryptocurrency at date $t + \Delta$ becomes:

$$P_{t+\Delta} = (1 - \Delta)P_t + \Delta P_{t+1}. \tag{A.4}$$

From (A.4), the return on the cryptocurrency over the period $[t, t + \Delta]$ is

$$r_{[t, t+\Delta]} = \frac{P_{t+\Delta}}{P_t} - 1 = \frac{\Delta(P_{t+1} - P_t)}{P_t} = \Delta r_{t+1},$$

while the return on the period $[t + \Delta, t + 1]$ is

$$r_{[t+\Delta, t+1]} = \frac{P_{t+1}}{P_{t+\Delta}} - 1 = \frac{(1 - \Delta)(P_{t+1} - P_t)/P_t}{[(1 - \Delta)P_t + \Delta P_{t+1}]/P_t} = \frac{(1 - \Delta)r_{t+1}}{1 + \Delta r_{t+1}}.$$

We assume that the first opening execution date is delayed by $\Delta \times 5$ minutes from the decision date $t = T$ and at each switching execution date d_j , $j = 2, \dots, L$, the position closing is delayed by $\Delta \times 5$ minutes and the opening of the new position by $2\Delta \times 5$ minutes from the decision date $t = d_j$, with $\Delta \in [0, 0.5]$.

In the presence of delays, we adjust the gross returns in equations (5.1)-(5.2). Concretely, for $\tilde{\Delta} \in [0, 1]$ over the period $[d_j + \tilde{\Delta}, t]$ with $t \leq d_{j+1}$ the delayed gross return of a buy order executed at $d_j + \tilde{\Delta}$ with an hourly rollover fee $\tau^R \in [0, 1]$ is:

$$B_t^j(\tilde{\Delta}) = 1 + R_t^j(\tilde{\Delta}) = (1 + r_{[d_j + \tilde{\Delta}, d_{j+1}]}) \prod_{t'=d_j+2}^t (1 + r_{t'}) \kappa(t', d_j), \quad (\text{A.5})$$

where $\kappa(t', d_j) = 1 - \tau^R 1_{\{t' \equiv d_j \pmod{12}\}}$. Suppose now that the holding period planned for the period $[d_j, d_{j+1}]$ corresponds to a short position. Since the cumulative return of a short position over the period $[d_j + \tilde{\Delta}, t]$ is the negative of the cumulative return on a long position over the same period, the gross return on the short position over the effective period $[d_j + \tilde{\Delta}, t]$ is:

$$D_t^j(\tilde{\Delta}) = 1 - R_t^j(\tilde{\Delta}) = 2 - B_t^j(\tilde{\Delta}) = 2 - (1 + r_{[d_j + \tilde{\Delta}, d_{j+1}]}) \prod_{t'=d_j+2}^t (1 + r_{t'}) \kappa(t', d_j). \quad (\text{A.6})$$

At the first trading date $t = T$, the capital is set to $C_T(\Delta) = 1$. The capital $C_t(\Delta)$ of the delayed trading at the exact dates $t = T + 1, \dots, T'$ using the DSR-MSM trading strategy can be expressed in the following recursive way. For a given $t = T + 1, \dots, T'$, let j be such that $d_j < t \leq d_{j+1}$, $j = 1, \dots, L$. Then $C_t(\Delta)$ is defined in the following way. At date $t = T + 1 = d_1 + 1$:

$$C_{T+1}(\Delta) = \begin{cases} (1 - \tau^O)(1 + r_{[T+\Delta, T+1]}) & \text{if } O_{T+1} = 1 \\ (1 - \tau^O)(1 - r_{[T+\Delta, T+1]}) & \text{if } O_{T+1} = -1 \end{cases}. \quad (\text{A.7})$$

For $d_1 + 1 < t \leq d_2$:

$$C_t(\Delta) = \begin{cases} C_{t-1}(\Delta)(1+r_t)\kappa(t, d_1 + \Delta) & \text{if } O_t = 1 \\ (1 - \tau^O)C_{d_1}(\Delta)D_t^1(\Delta) & \text{if } O_t = -1 \end{cases}. \quad (\text{A.8})$$

For $t = d_j + 1, j = 2, \dots, L$:

$$C_{d_j+1}(\Delta) = \begin{cases} (1 - \tau^O)^2 C_{d_{j-1}}(\Delta) [2 - B_{d_j}^{j-1}(2^{1_{\{j \neq 2\}}} \Delta) \kappa^\Delta(d_j, d_{j-1})] (1 + r_{[d_j+2\Delta, d_j+1]}) & \text{if } O_t = 1 \\ (1 - \tau^O) C_{d_j}(\Delta) \kappa^\Delta(d_j, d_{j-1}) (1 - r_{[d_j+2\Delta, d_j+1]}) & \text{if } O_t = -1 \end{cases}, \quad (\text{A.9})$$

where $\kappa^\Delta(d_j, d_{j-1}) = \kappa(d_j, d_{j-1} + \Delta)(1 + r_{[d_j, d_j+\Delta]})$.

For $d_j + 1 < t \leq d_{j+1}, j = 2, \dots, L$:

$$C_t(\Delta) = \begin{cases} C_{t-1}(\Delta)(1+r_t)\kappa(t, d_j + 2\Delta) & \text{if } O_t = 1 \\ (1 - \tau^O)C_{d_j}D_t^j(2\Delta) & \text{if } O_t = -1 \end{cases}. \quad (\text{A.10})$$

A.5 Additional Tables

Table A1

Parameter Estimates of the MSM Model via AML

	Frequency	$\mu \times 10^4$	$\nu \times 10$	$\rho \times 10$	m_0	$\bar{\gamma}$	b	$\sigma \times 10^2$	$\bar{k}$
BTC	5 min	0.113 (0.010)	0.668 (0.047)	1.182 (0.018)	1.360 (0.022)	0.901 (0.112)	1.751 (0.234)	0.201 (0.052)	23 (3.258)
	10 min	0.215 (0.022)	0.124 (0.113)	0.801 (0.024)	1.367 (0.019)	0.900 (0.127)	1.751 (0.239)	0.297 (0.055)	20 (2.441)
	15 min	0.351 (0.038)	-0.138 (0.347)	0.420 (0.031)	1.311 (0.022)	0.912 (0.180)	1.465 (0.204)	0.408 (0.073)	28 (4.593)
DASH	5 min	0.048 (0.064)	0.106 (0.586)	0.270 (0.017)	1.582 (0.010)	0.921 (0.166)	5.745 (0.533)	0.396 (0.015)	6 (0.331)
	10 min	0.030 (0.212)	-0.457 (0.208)	0.167 (0.024)	1.573 (0.013)	0.903 (0.219)	5.199 (0.634)	0.572 (0.025)	6 (0.425)
	15 min	0.019 (0.154)	-0.697 (0.231)	0.141 (0.028)	1.577 (0.018)	0.800 (0.184)	4.957 (0.715)	0.722 (0.037)	6 (0.619)
ETH	5 min	0.076 (0.025)	0.462 (0.226)	1.176 (0.017)	1.501 (0.028)	0.400 (0.055)	3.501 (0.856)	0.339 (0.022)	8 (1.208)
	10 min	0.117 (0.062)	0.026 (0.397)	0.646 (0.024)	1.505 (0.017)	0.583 (0.063)	3.366 (0.535)	0.508 (0.038)	8 (0.770)
	15 min	0.145 (0.069)	-0.361 (0.094)	0.407 (0.028)	1.502 (0.022)	0.696 (0.095)	3.183 (0.580)	0.619 (0.044)	8 (0.954)
LTC	5 min	-0.033 (0.057)	-0.899 (0.195)	0.418 (0.016)	1.323 (0.013)	0.954 (0.140)	1.721 (0.161)	0.354 (0.042)	21 (1.991)
	10 min	-0.086 (0.044)	-1.209 (0.191)	0.074 (0.027)	1.353 (0.013)	0.936 (0.111)	1.736 (0.153)	0.485 (0.045)	18 (1.428)
	15 min	-0.069 (0.079)	-1.481 (0.185)	-0.006 (0.041)	1.369 (0.013)	0.896 (0.115)	1.749 (0.161)	0.630 (0.061)	17 (1.571)
XRP	5 min	-0.043 (0.157)	-0.500 (1.699)	0.220 (0.029)	1.390 (0.033)	0.477 (0.062)	2.052 (0.414)	0.457 (0.134)	18 (3.326)
	10 min	-0.082 (1.051)	-0.676 (1.821)	-0.055 (0.255)	1.323 (0.030)	0.399 (0.095)	1.490 (0.288)	0.691 (0.162)	27 (6.156)
	15 min	-0.149 (0.073)	-0.859 (0.411)	-0.151 (0.039)	1.332 (0.039)	0.501 (0.069)	1.497 (0.069)	1.022 (0.310)	28 (7.684)

This table presents parameter estimates of the MSM model via AML for returns captured at five-minute, ten-minute, and fifteen-minute frequencies. Asymptotic standard errors are reported in parentheses.

A.6 Additional Figures

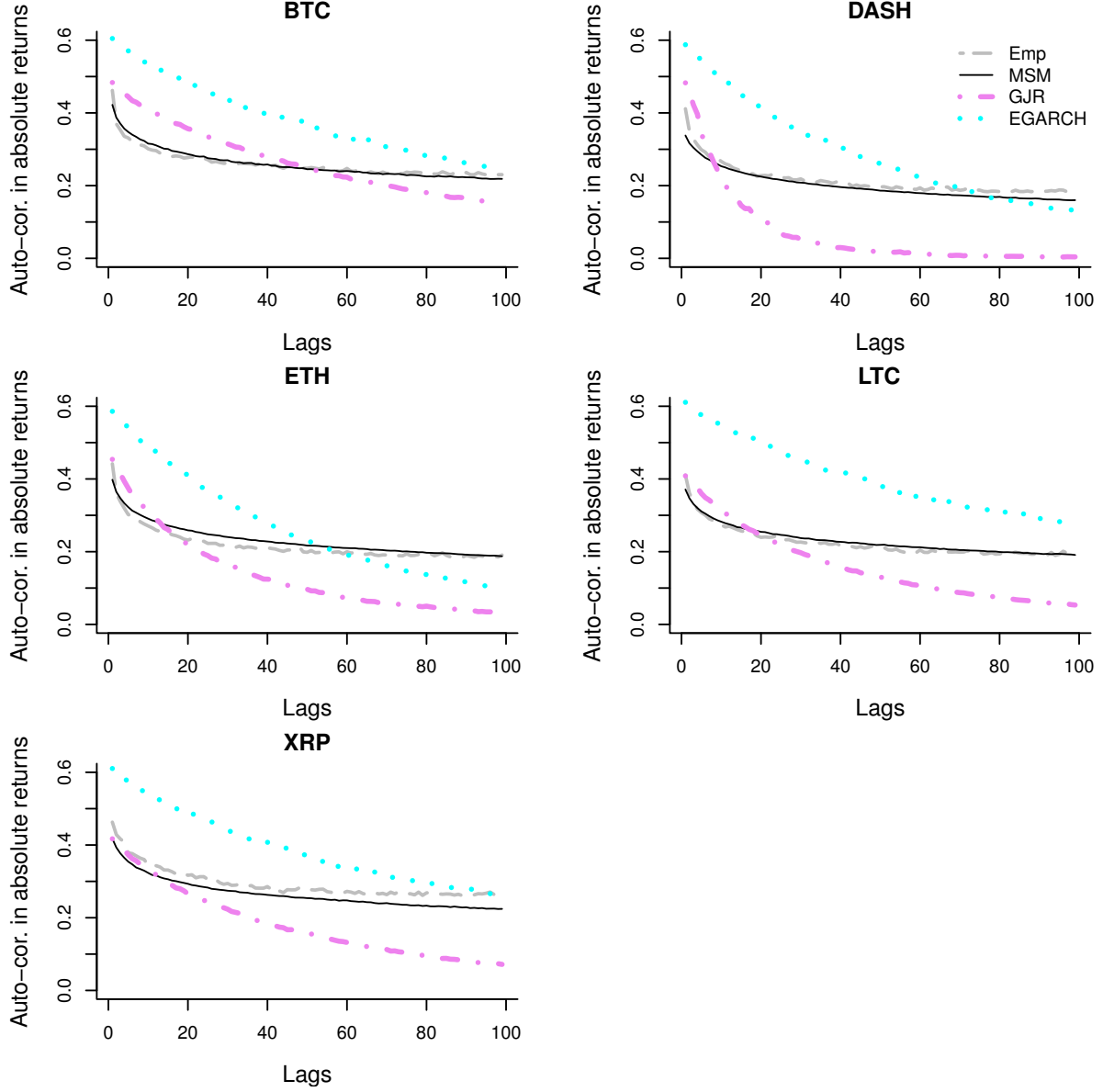


Fig. A1. *Autocorrelations in Absolute Returns.* Notes: This figure illustrates the autocorrelations in absolute returns for up to 100 lags in simulated and empirical (dashed lines) series for each cryptocurrency. Simulated series are generated from the estimated MSM (continuous lines), EGARCH (dotted lines), and GJR (dash-dotted lines) models. In simulated series, each reported AC_p , $p = 1, \dots, 100$, corresponds to the median of the AC_p values obtained in 100 simulated series.